

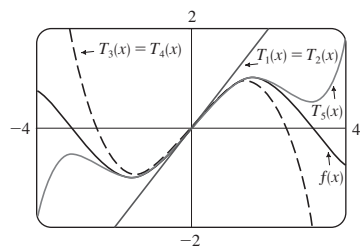
41. $T_1(x) = x$

$T_2(x) = x$

$T_3(x) = x - \frac{1}{6}x^3$

$T_4(x) = x - \frac{1}{6}x^3$

$T_5(x) = x - \frac{1}{6}x^3 + \frac{1}{120}x^5$



43. (a) $T_1(s) = 1 + 2s$ (b) $T_2(s) = 1 + 2s - 4s^2$

CHAPTER 3 REVIEW | page 240

True-False Quiz

1. False 3. True 5. True 7. False 9. False

11. True 13. True

Exercises

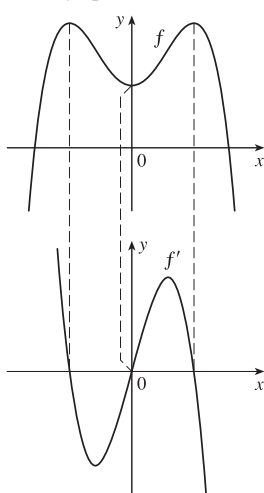
1. $f''(5)$, 0, $f'(5)$, $f'(2)$, 1, $f'(3)$

3. (a) The rate at which the cost changes with respect to the interest rate; dollars/(percent per year)

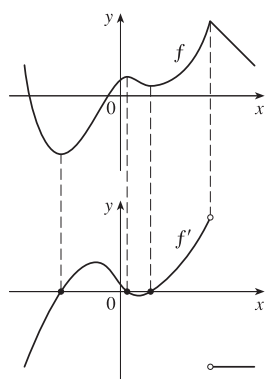
(b) As the interest rate increases past 10%, the cost is increasing at a rate of \$1200/(percent per year).

(c) Always positive

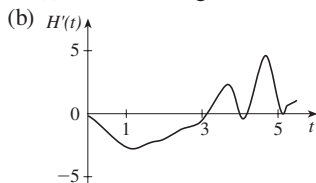
5.



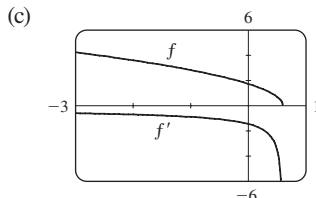
7.



9. (a) Rate of change of heart rate with respect to time.



11. (a) $f'(x) = -\frac{5}{2}(3 - 5x)^{-1/2}$ (b) $(-\infty, \frac{3}{5}]$, $(-\infty, \frac{3}{5})$



13. -4 (discontinuity), -1 (corner), 2 (discontinuity), 5 (vertical tangent)

15. $6x(x^4 - 3x^2 + 5)^2(2x^2 - 3)$

17. $\frac{1}{2\sqrt{x}} - \frac{4}{3\sqrt[3]{x^7}}$ 19. $\frac{2(2x^2 + 1)}{\sqrt{x^2 + 1}}$

21. $2 \cos 2\theta e^{\sin 2\theta}$

23. $\frac{t^2 + 1}{(1 - t^2)^2}$ 25. $-\frac{e^{1/x}(1 + 2x)}{x^4}$ 27. $\frac{1 - y^4 - 2xy}{4xy^3 + x^2 - 3}$

29. $\frac{2 \sec 2\theta (\tan 2\theta - 1)}{(1 + \tan 2\theta)^2}$ 31. $(1 + c^2)e^{cx} \sin x$

33. $\frac{2}{(1 + 2x) \ln 5}$ 35. $\frac{2x - y \cos(xy)}{x \cos(xy) + 1}$

37. $3^{x \ln x} (\ln 3)(1 + \ln x)$ 39. $\cot x - \sin x \cos x$

41. $\frac{4x}{1 + 16x^2} + \tan^{-1}(4x)$ 43. $5 \sec 5x$

45. $2 \cos \theta \tan(\sin \theta) \sec^2(\sin \theta)$

47. $\cos(\tan \sqrt{1 + x^3})(\sec^2 \sqrt{1 + x^3}) \frac{3x^2}{2\sqrt{1 + x^3}}$

49. $\frac{-3 \sin(e^{\sqrt{\tan 3x}}) e^{\sqrt{\tan 3x}} \sec^2(3x)}{2\sqrt{\tan 3x}}$

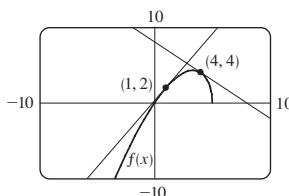
51. $-\frac{4}{27}$ 53. $2^x (\ln 2)^n$

55. $y = 2\sqrt{3}x + 1 - \pi\sqrt{3}/3$

57. $y = -x + 2$; $y = x + 2$

59. (a) $\frac{10 - 3x}{2\sqrt{5 - x}}$ (b) $y = \frac{7}{4}x + \frac{1}{4}$, $y = -x + 8$

(c)



61. (a) 2 (b) 44

63. $2xg(x) + x^2g'(x)$ 65. $2g(x)g'(x)$

67. $g'(e^x)e^x$ 69. $g'(x)/g(x)$ 71. $(-3, 0)$

73. $(\pm 2/\sqrt{6}, \mp 1/\sqrt{6})$

75. $v(t) = -Ae^{-ct}[c \cos(\omega t + \delta) + \omega \sin(\omega t + \delta)]$,
 $a(t) = Ae^{-ct}[(c^2 - \omega^2) \cos(\omega t + \delta) + 2c\omega \sin(\omega t + \delta)]$

77. (a) $dV/dh = \frac{1}{3}\pi r^2$ (b) $dV/dr = \frac{2}{3}\pi rh$

79. 1980: $E'(10) \approx \$21.56$ billion/year;
2000: $E'(30) \approx \$125.64$ billion/year

81. (a) $200(3.24)^t$ (b) $\approx 22,040$

(c) $\approx 25,910$ bacteria/h (d) $(\ln 50)/(\ln 3.24) \approx 3.33$ h

83. (a) $C_0 e^{-kt}$ (b) ≈ 100 h

85. $\frac{4}{3} \text{ cm}^2/\text{min}$ 87. 1.297383

89. (a) $L(x) = 1 + x$; $\sqrt[3]{1 + 3x} \approx 1 + x$; $\sqrt[3]{1.03} \approx 1.01$
(b) $-0.235 < x < 0.401$

91. (a) $T_1(c) = \ln 2 + 2(c - 1)$

(b) $T_2(c) = \ln 2 + 2(c - 1) - 4(c - 1)^2$

93. $(\cos \theta)'|_{\theta=\pi/3} = -\sqrt{3}/2$

CHAPTER 4

■ EXERCISES 4.1 | page 256

Abbreviations: abs, absolute; loc, local; max, maximum; min, minimum

1. Abs min: smallest function value on the entire domain of the function; loc min at c : smallest function value when x is near c

3. Abs max at s , abs min at r , loc max at c , loc min at b and r

5. Local max: $f(0.18) \approx 0.36$ mV, $f(0.30) \approx 1.2$ mV, $f(0.57) \approx 0.31$ mV

Abs max: $f(0.30) \approx 1.2$ mV

Local min: $f(0.29) \approx -0.01$ mV, $f(0.32) \approx -0.76$ mV

Abs Min: $f(0.32) \approx -0.76$ mV

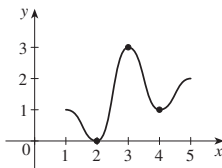
7. Local max: $D(\text{Oct } 23, 1918) \approx 91$, $D(\text{Jan } 22, 1919) \approx 15$

Abs max: $D(\text{Oct } 23, 1918) \approx 91$

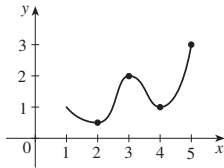
Local min: $D(\text{Nov } 27, 1918) \approx 4$, $D(\text{Feb } 13, 1919) \approx 7$

Abs min: $D(\text{Sept } 11, 1918) \approx 0$

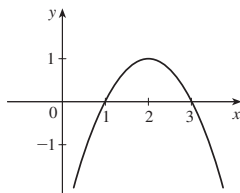
9.



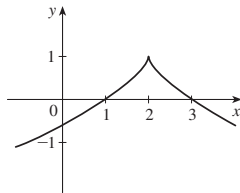
11.



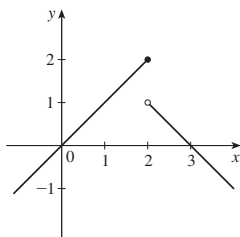
13. (a)



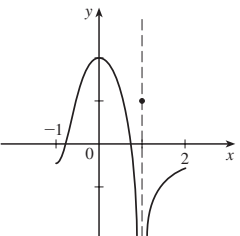
(b)



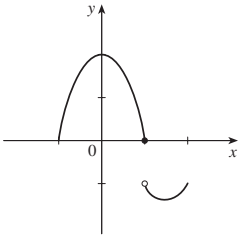
(c)



15. (a)



(b)



17. Abs max $f(3) = 4$ 19. None

21. Abs max $f(2) = \ln 2$ 23. Abs max $f(0) = 1$

25. $\frac{1}{3}$ 27. $-4, 2$

29. $0, \frac{1}{2}(-1 \pm \sqrt{5})$ 31. $0, 2$

33. $0, \frac{4}{9}$ 35. $0, \frac{8}{7}, 4$ 37. $n\pi$ (n an integer) 39. $0, \frac{2}{3}$

41. $f(2) = 16, f(5) = 7$ 43. $f(-1) = 8, f(2) = -19$

45. $f(3) = 66, f(\pm 1) = 2$ 47. $f(\sqrt{2}) = 2, f(-1) = -\sqrt{3}$

49. $f(2) = 2/\sqrt[3]{e}, f(-1) = -1/\sqrt[3]{e}$

51. $f(1) = \ln 3, f(-\frac{1}{2}) = \ln \frac{3}{4}$

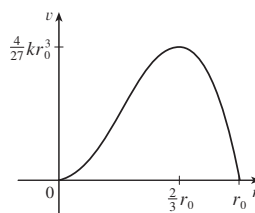
53. $f(\pi/6) = \frac{3}{2}\sqrt{3}, f(\pi/2) = 0$

55. $f\left(\frac{a}{a+b}\right) = \frac{a^a b^b}{(a+b)^{a+b}}$

57. Smallest: $P(2.16) \approx 5.32$; largest: $P(0) = P(9) = 10$

59. (a) $r = \frac{2}{3}r_0$ (b) $v = \frac{4}{27}kr_0^3$

(c)



61. $\approx 3.9665^\circ\text{C}$

■ EXERCISES 4.2 | page 271

Abbreviations: inc, increasing; dec, decreasing; CD, concave downward; CU, concave upward; HA, horizontal asymptote; VA, vertical asymptote; IP, inflection point(s)

1. 0.8, 3.2, 4.4, 6.1

5. (a) (1, 3), (4, 6) (b) (0, 1), (3, 4) (c) (0, 2)

(d) (2, 4), (4, 6) (e) (2, 3)

7. (a) See the First Derivative Test

(b) See the Second Derivative Test and the note that precedes Example 9.

9. (a) 3, 5 (b) 2, 4, 6 (c) 1, 7

11. (a) Inc on $(-\infty, -3)$, $(2, \infty)$; dec on $(-3, 2)$

(b) Loc max $f(-3) = 81$; loc min $f(2) = -44$

(c) CU on $(-\frac{1}{2}, \infty)$; CD on $(-\infty, -\frac{1}{2})$; IP $(-\frac{1}{2}, \frac{37}{2})$

13. (a) Inc on $(-1, 0)$, $(1, \infty)$; dec on $(-\infty, -1)$, $(0, 1)$

(b) Loc max $f(0) = 3$; loc min $f(\pm 1) = 2$

(c) CU on $(-\infty, -\sqrt{3}/3)$, $(\sqrt{3}/3, \infty)$;
CD on $(-\sqrt{3}/3, \sqrt{3}/3)$; IP $(\pm\sqrt{3}/3, \frac{22}{9})$

15. (a) Inc on $(0, \pi/4)$, $(5\pi/4, 2\pi)$; dec on $(\pi/4, 5\pi/4)$

(b) Loc max $f(\pi/4) = \sqrt{2}$; loc min $f(5\pi/4) = -\sqrt{2}$

(c) CU on $(3\pi/4, 7\pi/4)$; CD on $(0, 3\pi/4)$, $(7\pi/4, 2\pi)$;
IP $(3\pi/4, 0)$, $(7\pi/4, 0)$

17. (a) Inc on $(-\frac{1}{3} \ln 2, \infty)$; dec on $(-\infty, -\frac{1}{3} \ln 2)$

(b) Loc min $f(-\frac{1}{3} \ln 2) = 2^{-2/3} + 2^{1/3}$ (c) CU on $(-\infty, \infty)$

19. (a) Inc on $(0, e^2)$; dec on (e^2, ∞)

(b) Loc max $f(e^2) = 2/e$

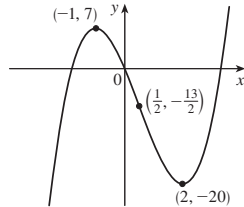
(c) CU on $(e^{8/3}, \infty)$; CD on $(0, e^{8/3})$; IP $(e^{8/3}, \frac{8}{3}e^{-4/3})$

21. Loc max $f(\frac{3}{4}) = \frac{5}{4}$

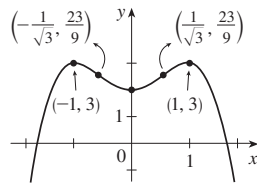
23. (a) f has a local maximum at 2.

(b) f has a horizontal tangent at 6.

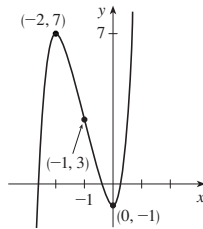
25. (a) Inc on $(-\infty, -1)$, $(2, \infty)$;
dec on $(-1, 2)$
(b) Loc max $f(-1) = 7$;
loc min $f(2) = -20$
(c) CU on $(\frac{1}{2}, \infty)$; CD on $(-\infty, \frac{1}{2})$;
IP $(\frac{1}{2}, -\frac{13}{2})$
(d) See graph at right.



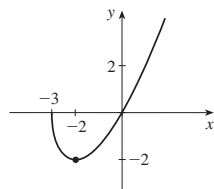
27. (a) Inc on $(-\infty, -1)$, $(0, 1)$;
dec on $(-1, 0)$, $(1, \infty)$
(b) Loc max $f(-1) = 3$, $f(1) = 3$;
loc min $f(0) = 2$
(c) CU on $(-1/\sqrt{3}, 1/\sqrt{3})$;
CD on $(-\infty, -1/\sqrt{3})$, $(1/\sqrt{3}, \infty)$;
IP $(\pm 1/\sqrt{3}, \frac{23}{9})$
(d) See graph at right.



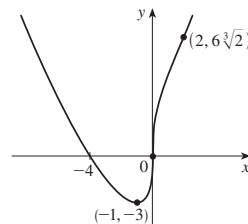
29. (a) Inc on $(-\infty, -2)$, $(0, \infty)$;
dec on $(-2, 0)$
(b) Loc max $h(-2) = 7$;
loc min $h(0) = -1$
(c) CU on $(-1, \infty)$;
CD on $(-\infty, -1)$; IP $(-1, 3)$
(d) See graph at right.



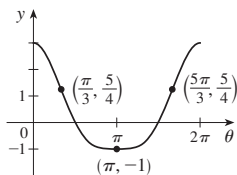
31. (a) Inc on $(-2, \infty)$; dec on $(-3, -2)$
(b) Loc min $A(-2) = -2$
(c) CU on $(-3, \infty)$
(d) See graph at right.



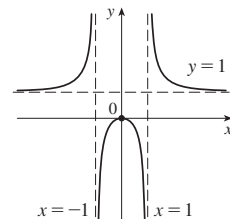
33. (a) Inc on $(-1, \infty)$;
dec on $(-\infty, -1)$
(b) Loc min $C(-1) = -3$
(c) CU on $(-\infty, 0)$, $(2, \infty)$;
CD on $(0, 2)$;
IP $(0, 0)$, $(2, 6\sqrt[3]{2})$
(d) See graph at right.



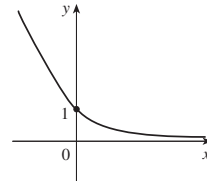
35. (a) Inc on $(\pi, 2\pi)$;
dec on $(0, \pi)$
(b) Loc min $f(\pi) = -1$
(c) CU on $(\pi/3, 5\pi/3)$;
CD on $(0, \pi/3)$, $(5\pi/3, 2\pi)$;
IP $(\pi/3, \frac{5}{4})$, $(5\pi/3, \frac{5}{4})$
(d) See graph at right.



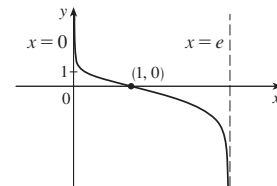
37. (a) HA $y = 1$, VA $x = -1$, $x = 1$
(b) Inc on $(-\infty, -1)$, $(-1, 0)$;
dec on $(0, 1)$, $(1, \infty)$
(c) Loc max $f(0) = 0$
(d) CU on $(-\infty, -1)$, $(1, \infty)$;
CD on $(-1, 1)$
(e) See graph at right.



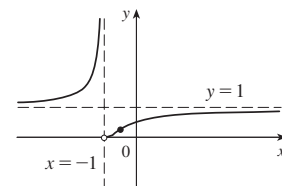
39. (a) HA $y = 0$
(b) Dec on $(-\infty, \infty)$
(c) None
(d) CU on $(-\infty, \infty)$
(e) See graph at right.



41. (a) VA $x = 0$, $x = e$
(b) Dec on $(0, e)$
(c) None
(d) CU on $(0, 1)$; CD on $(1, e)$;
IP $(1, 0)$
(e) See graph at right.



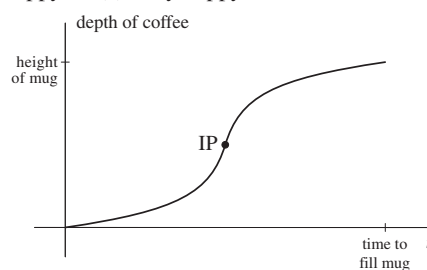
43. (a) HA $y = 1$, VA $x = -1$
(b) Inc on $(-\infty, -1)$, $(-1, \infty)$
(c) None
(d) CU on $(-\infty, -1)$, $(-1, -\frac{1}{2})$;
CD on $(-\frac{1}{2}, \infty)$;
IP $(-\frac{1}{2}, 1/e^2)$
(e) See graph at right.



45. (3, ∞)

47. (a) Very unhappy (b) Unhappy
(c) Happy (d) Very happy

49.

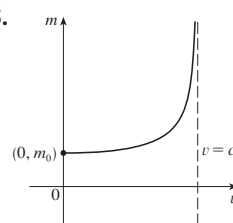


51. 28.57 min, when the rate of increase of drug level in the bloodstream is greatest; 85.71 min, when rate of decrease is greatest

53. (a) $c = 1$

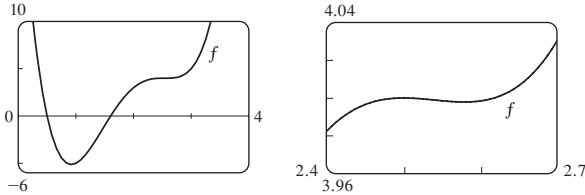
- (b) Constant concentration treatment yields a better response.

55.

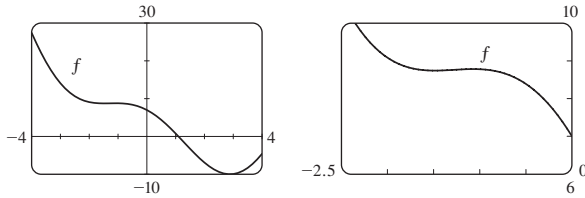


57. $f(x) = \frac{1}{9}(2x^3 + 3x^2 - 12x + 7)$

59. Inc on $(0.92, 2.5)$, $(2.58, \infty)$; dec on $(-\infty, 0.92)$, $(2.5, 2.58)$;
 loc max $f(2.5) = 4$; loc min $f(0.92) \approx -5.12$, $f(2.58) \approx 3.998$;
 CU on $(-\infty, 1.46)$, $(2.54, \infty)$; CD on $(1.46, 2.54)$;
 IP $(1.46, -1.40)$, $(2.54, 3.999)$

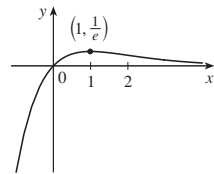


61. Inc on $(-1.49, -1.07)$, $(2.89, 4)$;
 dec on $(-4, -1.49)$, $(-1.07, 2.89)$;
 loc max $f(-1.07) \approx 8.79$;
 loc min $f(-1.49) \approx 8.75$, $f(2.89) \approx -9.99$;
 CU on $(-4, -1.28)$, $(1.28, 4)$; CD on $(-1.28, 1.28)$;
 IP $(-1.28, 8.77)$, $(1.28, -1.48)$

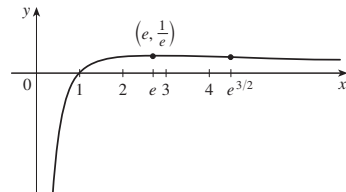


EXERCISES 4.3 | page 282

1. 2 3. $-\infty$ 5. ∞ 7. 0 9. $-\infty$ 11. 3 13. $\ln \frac{5}{3}$
 15. $\frac{1}{2}$ 17. $-1/\pi^2$ 19. $\frac{1}{2}a(a-1)$ 21. $\frac{1}{24}$ 23. π
 25. 3 27. 0 29. $\frac{1}{2}$ 31. $\frac{1}{2}$ 33. ∞
 35. HA $y = 0$



37. HA $y = 0$, VA $x = 0$



39. Fastest to slowest: $y = e^{3x}$, $y = e^{2x}$, $y = x^5$, $y = \ln(x^{10})$

41. Fastest to slowest: $y = \sqrt{x}$, $y = \sqrt[3]{x}$, $y = (\ln x)^3$, $y = (\ln x)^2$

43. ∞

45. Fastest to slowest: $y = e^{-x}$, $y = \frac{1}{x^2}$, $y = \frac{1}{x}$, $y = x^{-1/2}$

47. 1

49. (a) r (b) r (c) $f'(0)$

51. (a) 62.3%

(b) 80.1%; Answer is larger due to Stiles-Crawford effect. Light entering closer to center of pupil measures brighter.

(c) $\lim_{r \rightarrow 0} P = 100\%$;

Result expected: all light entering at center of pupil is sensed at retina.

53. $te^{-\beta t}$ 57. $\frac{16}{9}a$

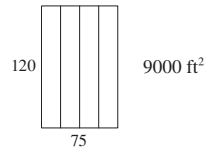
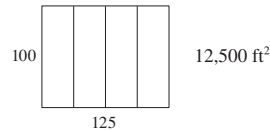
EXERCISES 4.4 | page 293

1. (a) 11, 12 (b) 11.5, 11.5

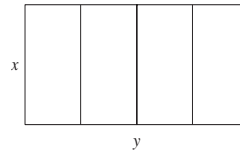
3. 10, 10 5. 25 m by 25 m

7. $N = 1$

9. (a)



(b)



(c) $A = xy$ (d) $5x + 2y = 750$ (e) $A(x) = 375x - \frac{5}{2}x^2$

(f) 14,062.5 ft²

11. 4000 cm³ 15. $(-\frac{6}{5}, \frac{3}{5})$

17. (a) $a = e$ (b) $a = 1/\mu$

19. (b) (i) \$342,491; \$342/unit; \$390/unit (ii) 400
 (iii) \$320/unit

21. (a) $h = r/2$

(b) $h = \frac{r}{2} + \frac{\alpha}{2p}$; $N = \frac{K}{2} \left(1 - \frac{\alpha}{rp} \right)$

(c) Part (b) gives a larger harvest and smaller stabilized fish population. Extra harvesting results in reduced unit costs making it profitable to harvest more.

23. (a) Function starts at zero and increases with time; concave down implies diminishing returns.

27. (a) $\frac{3}{2}s^2 \csc \theta (\csc \theta - \sqrt{3} \cot \theta)$

(b) $\cos^{-1}(1/\sqrt{3}) \approx 55^\circ$ (c) $6s[h + s/(2\sqrt{2})]$

29. (a) About 5.1 km from B

(b) C is close to B; C is close to D;

$W/L = \sqrt{25 + x^2}/x$, where $x = |BC|$

(c) ≈ 1.07 ; no such value (d) $\sqrt{41}/4 \approx 1.6$

■ EXERCISES 4.5 | page 305

1. Stable: $x = 1$ 3. Stable: $x = 1$; Unstable $x = 0, 2$

5. Stable: $x = 0$; Unstable: $x = 2$

7. Stable: $x = 0.8$; Unstable: $x = 0$

9. Unstable: $x = 0, \frac{1}{2}\ln(10)$

11. Stable: $x = 0, 3$; Unstable: $x = 1$

$\lim_{t \rightarrow \infty} x_t = 0$ when $x_0 = 0.5$

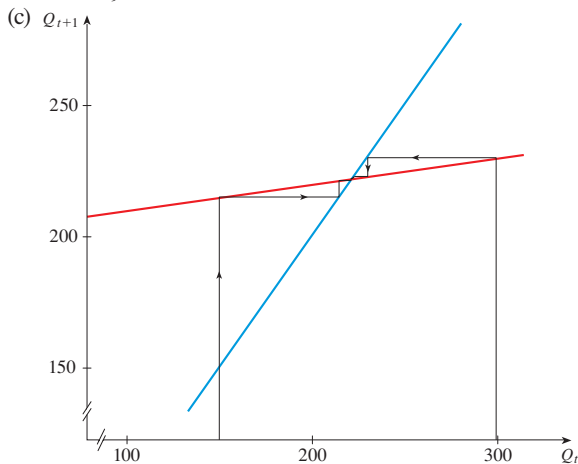
$\lim_{t \rightarrow \infty} x_t = 3$ when $x_0 = 2$

13. $x = 0$ is stable equilibrium when $|c| < 1$.

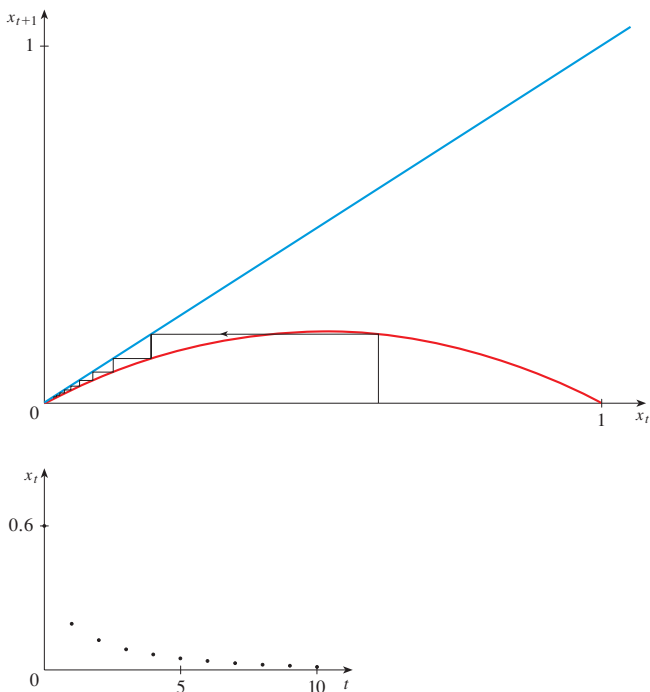
$x = c - 1$ is stable equilibrium when $c < -1$ or $c > 1$.

15. (a) $Q_{n+1} = 0.1Q_n + 200$

(b) $Q = \frac{2000}{9} = 222.\bar{2}$

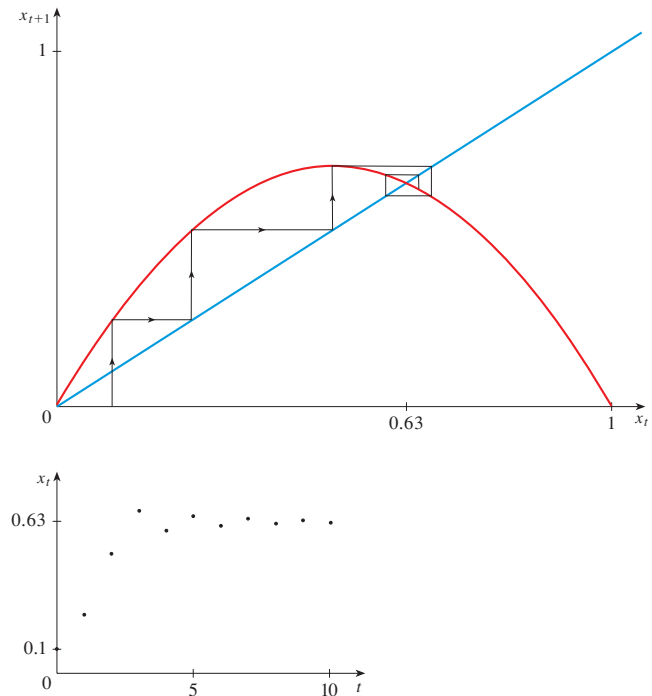


17.



$\hat{x} = 0$ is a stable equilibrium.

19.



$\hat{x} = \frac{17}{27} = 0.\overline{629}$ is a stable equilibrium; $\hat{x} = 0$ is unstable.

21. $N = 0$ is a stable equilibrium when $r < h$.

$N = K\left(1 - \frac{h}{r}\right)$ is a stable equilibrium when $h < r < h + 2$.

23. $d = 1$ is a stable equilibrium.

25. (a) $R = \frac{\theta K}{\theta + d}$ is a stable equilibrium when $0 < d + \theta < 2$.

(b) $R = 0$ is an unstable equilibrium;

$R = \sqrt{\frac{a - db}{d}}$ is a stable equilibrium.

■ EXERCISES 4.6 | page 311

1. $F(x) = \frac{1}{2}x + \frac{1}{4}x^3 - \frac{1}{5}x^4 + C$

3. $F(x) = \frac{2}{3}x^3 + \frac{1}{2}x^2 - x + C$

5. $F(x) = 4x^{5/4} - 4x^{7/4} + C$ 7. $F(x) = 4x^{3/2} - \frac{6}{7}x^{7/6} + C$

9. $F(x) = \sqrt{2}x + C$

11. $C(t) = -\frac{3}{t} + K$, where K is a constant.

13. $G(\theta) = \sin \theta + 5 \cos \theta + C$ 15. $V(s) = 2s^2 + 3e^s + C$

17. $F(u) = \frac{1}{3}u^3 - 6u^{-1/2} + C$ 19. $F(t) = \frac{1}{3}t^3 - t - \frac{1}{t} + C$

21. $y(t) = \frac{1}{3}t^3 + t + 6, t \geq 0$ 23. $P(t) = \frac{2}{3}e^{3t} + \frac{1}{3}, t \geq 0$

25. $r(\theta) = \sin \theta + \sec \theta + 2 - \frac{\sqrt{3}}{2}, 0 < \theta < \pi/2$

27. $u(t) = \frac{2}{3}t^{3/2} + 4t^{1/2} + \frac{1}{3}, t > 0$ 29. $x^3 + x^4 + Cx + D$

31. $\frac{3}{20}x^{8/3} + Cx + D$ 33. $x - 3x^2 + 8$

35. $4x^{3/2} + 2x^{5/2} + 4$

37. $-\sin \theta - \cos \theta + 5\theta + 4$

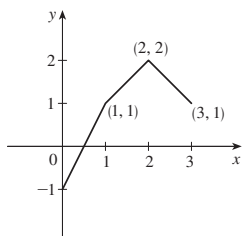
39. $x^2 - 2x^3 + 9x + 9$ 41. ≈ 44 bacteria

43. $s(t) = 1 - \cos t - \sin t$

45. (a) $s(t) = 450 - 4.9t^2$ (b) $\sqrt{450/4.9} \approx 9.58$ s

(c) $-9.8\sqrt{450/4.9} \approx -93.9$ m/s

47.



CHAPTER 4 REVIEW | page 312

True-False Quiz

1. False 3. False 5. True 7. False 9. True
11. True 13. False 15. True 17. True

Exercises

1. Abs max $f(4) = 5$, abs and loc min $f(3) = 1$

3. Abs max $f(2) = \frac{2}{5}$, abs and loc min $f(-\frac{1}{3}) = -\frac{9}{2}$

5. Abs max $f(\pi) = \pi$; abs min $f(0) = 0$;

loc max $f(\pi/3) = (\pi/3) + \frac{1}{2}\sqrt{3}$;

loc min $f(2\pi/3) = (2\pi/3) - \frac{1}{2}\sqrt{3}$

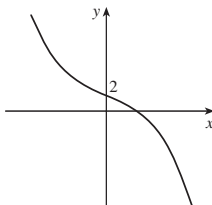
7. (a) None

(b) Dec on $(-\infty, \infty)$

(c) None

(d) CU on $(-\infty, 0)$; CD on $(0, \infty)$;IP $(0, 2)$

(e) See graph at right.



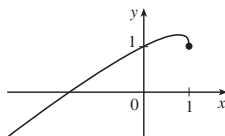
9. (a) None

(b) Inc on $(-\infty, \frac{3}{4})$; dec on $(\frac{3}{4}, 1)$

(c) Loc max $f(\frac{3}{4}) = \frac{5}{4}$

(d) CD on $(-\infty, 1)$;

(e) See graph at right.



11. (a) None

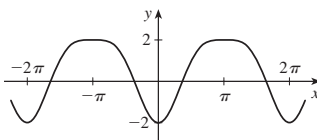
(b) Inc on $(2n\pi, (2n+1)\pi)$, n an integer;dec on $((2n+1)\pi, (2n+2)\pi)$

(c) Loc max $f((2n+1)\pi) = 2$

loc min $f(2n\pi) = -2$

(d) CU on $(2n\pi - (\pi/3), 2n\pi + (\pi/3))$;CD on $(2n\pi + (\pi/3), 2n\pi + (5\pi/3))$; IPs $(2n\pi \pm (\pi/3), -\frac{1}{4})$

(e) See graph at right.



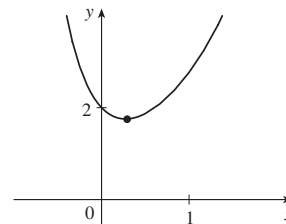
13. (a) None

(b) Inc on $(\frac{1}{4} \ln 3, \infty)$;dec on $(-\infty, \frac{1}{4} \ln 3)$

(c) Loc min $f(\frac{1}{4} \ln 3) = 3^{1/4} + 3^{-3/4}$

(d) CU on $(-\infty, \infty)$

(e) See graph at right.



15. (a) Max: $C(2.118) \approx 0.7567$ $\mu\text{g/mL}$

(b) IP: $t = \frac{5}{2} \ln(49/9) \approx 4.236$ h

Rate of change of concentration begins to increase after 4.236 h.

17. Population ≤ 900 after five weeks

19. π 21. 8 23. 0

25. $\frac{1}{2}$ 27. 500 and 125

29. $L = C$ 31. Δp is largest when $p = 1/2$

33. Stable: $x = 3/5$; unstable: $x = 0$

35. Stable: $x = 0, 4$; unstable: $x = 2$

$\lim_{t \rightarrow \infty} x_t = 0$ when $x_0 = 1$; $\lim_{t \rightarrow \infty} x_t = 4$ when $x_0 = 3$

37. $F(x) = -\cos x + \sec x + C, 0 \leq x \leq \pi/2$

39. $Q(t) = \frac{1}{4}t^4 + \frac{1}{3}t^3 - \frac{1}{2}t^2 + t + C$

41. $y(t) = t - \frac{1}{\pi}e^{\pi t} + \frac{1}{\pi}$

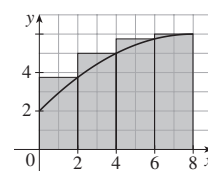
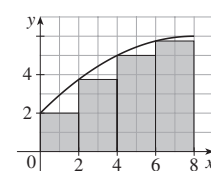
43. $f(x) = \frac{1}{2}x^2 - x^3 + 4x^4 + 2x + 1$

45. $s(t) = -\sin t - 3 \cos t + 3t + 3$

CHAPTER 5

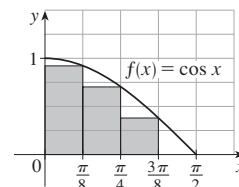
EXERCISES 5.1 | page 327

1. (a) $L_4 = 33, R_4 = 41$

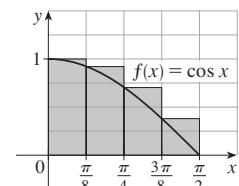


(b) $L_8 \approx 35.2, R_8 \approx 39.2$

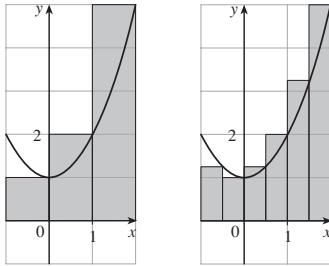
3. (a) 0.7908, underestimate



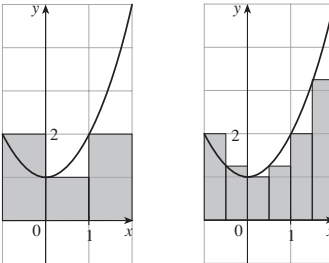
(b) 1.1835, overestimate



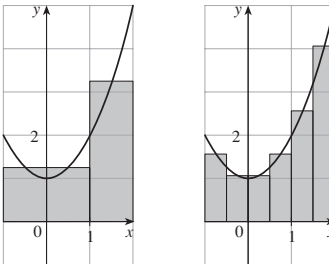
5. (a) 8, 6.875



(b) 5, 5.375



(c) 5.75, 5.9375

(d) M_6

7. 34.7 ft, 44.8 ft

9. $M_6 = 7840$ (infected cells/mL) · days11. $L_6 \approx 24$ people, $R_6 \approx 28$ people

13. 155 ft

$$15. \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2(1 + 2i/n)}{(1 + 2i/n)^2 + 1} \cdot \frac{2}{n} \quad 17. \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i\pi}{2n} \cos \frac{i\pi}{2n} \right) \frac{\pi}{2n}$$

19. (a) $L_n < A < R_n$

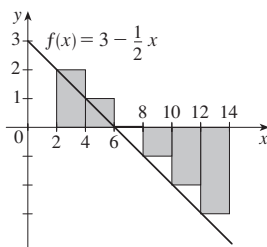
$$21. (a) \lim_{n \rightarrow \infty} \frac{64}{n^6} \sum_{i=1}^n i^5 \quad (b) \frac{n^2(n+1)^2(2n^2+2n-1)}{12} \quad (c) \frac{32}{3}$$

23. $\sin b, 1$

■ EXERCISES 5.2 | page 339

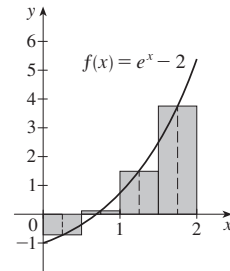
1. -6

The Riemann sum represents the sum of the areas of the two rectangles above the x -axis minus the sum of the areas of the three rectangles below the x -axis; that is, the net area of the rectangles with respect to the x -axis.



3. 2.322986

The Riemann sum represents the sum of the areas of the three rectangles above the x -axis minus the area of the rectangle below the x -axis.



5. (a) 4 (b) 6 (c) 10

7. Lower, $L_5 = -64$; upper, $R_5 = 16$ 9. 124.1644 11. 0.3084 13. 198 ($\mu\text{g/mL}$) · min

$$15. \int_2^6 x \ln(1 + x^2) dx \quad 17. \int_1^8 \sqrt{2x + x^2} dx \quad 19. 42$$

$$21. \frac{4}{3} \quad 23. 3.75 \quad 25. \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2 + 4i/n}{1 + (2 + 4i/n)^5} \cdot \frac{4}{n}$$

27. (a) 4 (b) 10 (c) -3 (d) 2

29. $-\frac{3}{4}$ 31. $3 + 9\pi/4$ 33. 2.5 35. 0

$$37. \int_{-1}^5 f(x) dx \quad 39. 122$$

41. $B < E < A < D < C$ 43. 15 47. $\int_0^1 x^4 dx$

■ EXERCISES 5.3 | page 350

$$1. -\frac{10}{3} \quad 3. \frac{56}{15} \quad 5. \frac{5}{9} \quad 7. -2 + 1/e \quad 9. \frac{49}{3} \quad 11. \frac{40}{3}$$

$$13. \frac{55}{63} \quad 15. 1 \quad 17. \ln 3 \quad 19. \frac{1}{e+1} + e - 1$$

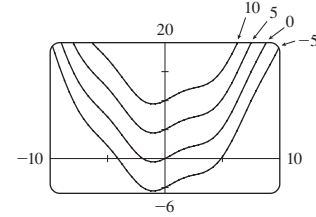
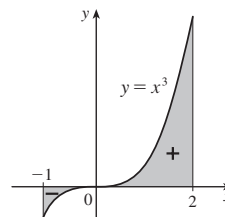
$$21. e^2 - 1 \quad 23. \ln 2 + 7 \quad 25. 1 + \pi/4 \quad 27. \pi/6$$

29. The function $f(x) = 1/x^2$ is not continuous on the interval $[-1, 3]$, so the Evaluation Theorem cannot be applied.

31. 2

33. 3.75

$$37. \sin x + \frac{1}{4}x^2 + C$$



$$39. 2t - t^2 + \frac{1}{3}t^3 - \frac{1}{4}t^4 + C \quad 41. \tan \alpha + C$$

$$43. \sec x + C \quad 45. 7848 \text{ (number of cells per mL)} \cdot \text{days}$$

47. The increase in the child's weight (in pounds) between the ages of 5 and 10

49. The total number of sea urchins between points a and b

51. The change in concentration between time t_1 and t_2

53. The number of gallons of oil leaked in the first two hours

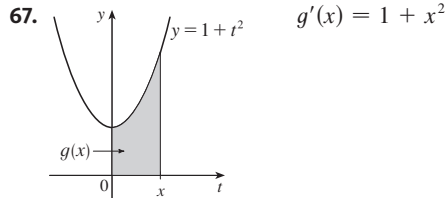
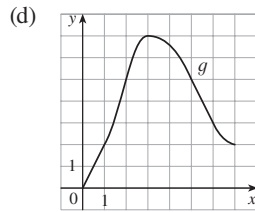
55. 1800 liters 57. 29.8 cm

$$59. (a) \int_0^L A(x) dx \quad (b) \text{Upper: } \beta L; \text{lower: } \alpha L$$

$$61. (a) P(t) = P_{\max}(1 - e^{-akt}) \quad (b) P_{\max} \left[5 + \frac{1}{ak} e^{-5ka} - \frac{1}{ak} \right]$$

$$(c) P_{\max} \left[t + \frac{1}{ak} e^{-akt} - \frac{1}{ak} \right] \quad (d) P_{\max}(1 - e^{-akt})$$

65. (a) 0, 2, 5, 7, 3
 (b) (0, 3)
 (c) $x = 3$



69. $g'(x) = 1/(x^3 + 1)$ 71. $g'(y) = y^2 \sin y$
 73. $F'(x) = -\sqrt{1 + \sec x}$ 75. $h'(x) = -\frac{\arctan(1/x)}{x^2}$
 77. $y' = \sqrt{\tan x + \sqrt{\tan x}} \sec^2 x$ 79. 29 81. 3 83. $\frac{1}{4}$
 85. $f(x) = x^{3/2}$, $a = 9$

EXERCISES 5.4 | page 360

1. $-e^{-x} + C$ 3. $\frac{2}{9}(x^3 + 1)^{3/2} + C$ 5. $-\frac{1}{4}\cos^4\theta + C$
 7. $-\frac{1}{2}\cos(x^2) + C$ 9. $\frac{1}{63}(3x - 2)^{21} + C$
 11. $-(1/\pi)\cos \pi t + C$ 13. $\frac{1}{3}(\ln x)^3 + C$
 15. $-\frac{1}{3}\ln|5 - 3x| + C$ 17. $\frac{2}{3}\sqrt{3ax + bx^3} + C$
 19. $\frac{2}{3}(1 + e^x)^{3/2} + C$ 21. $-1/(\sin x) + C$
 23. $\frac{1}{15}(x^3 + 3x)^5 + C$ 25. $-\frac{2}{3}(\cot x)^{3/2} + C$
 27. $-\frac{1}{2}\cos(e^{2x}) + C$ 29. $\frac{1}{3}\sec^3 x + C$
 31. $\frac{1}{40}(2x + 5)^{10} - \frac{5}{36}(2x + 5)^9 + C$ 33. $-\ln(1 + \cos^2 x) + C$
 35. $\tan^{-1}x + \frac{1}{2}\ln(1 + x^2) + C$ 37. $2/\pi$ 39. $\frac{45}{28}$ 41. $\frac{182}{9}$
 43. $2(e^2 - e)$ 45. 0 47. $\frac{16}{15}$ 49. $\ln(e + 1)$ 51. $\frac{1}{6}$
 53. 6π 55. All three areas are equal. 57. ≈ 4512 L
 59. $\frac{5}{4\pi}\left(1 - \cos \frac{2\pi t}{5}\right)$ L
 61. ≈ 1.0 mm³ 63. ≈ 515 degree-days 65. 5

EXERCISES 5.5 | page 367

1. $\frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C$ 3. $\frac{1}{5}x \sin 5x + \frac{1}{25} \cos 5x + C$
 5. $2(r - 2)e^{r/2} + C$
 7. $-\frac{1}{\pi}x^2 \cos \pi x + \frac{2}{\pi^2}x \sin \pi x + \frac{2}{\pi^3} \cos \pi x + C$
 9. $x \ln \sqrt[3]{x} - \frac{1}{3}x + C$ 11. $\frac{1}{13}e^{2\theta}(2 \sin 3\theta - 3 \cos 3\theta) + C$
 13. $\pi/3$ 15. $\frac{1}{2} - \frac{1}{2} \ln 2$ 17. $\frac{1}{4} - \frac{3}{4}e^{-2}$
 19. $2(\ln 2)^2 - 4 \ln 2 + 2$ 21. $2\sqrt{x} \sin \sqrt{x} + 2 \cos \sqrt{x} + C$
 23. $-\frac{1}{2} - \pi/4$ 25. $\frac{1}{2}(x^2 - 1) \ln(1 + x) - \frac{1}{4}x^2 + \frac{1}{2}x + \frac{3}{4} + C$
 27. (c) $-\frac{1}{4}\cos x \sin^3 x + \frac{3}{8}x - \frac{3}{16}\sin 2x + C$
 31. $x[(\ln x)^3 - 3(\ln x)^2 + 6 \ln x - 6] + C$
 33. ≈ 10.3560 ($\mu\text{g/mL}$) $\cdot$ h
 35. $\frac{1}{2}e^{-1} \cos 1 \approx 0.09938$ 37. 2

EXERCISES 5.6 | page 370

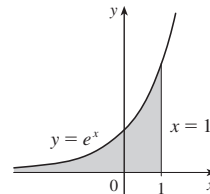
1. (a) $\frac{1}{x^2 - 1} = \frac{A}{x + 1} + \frac{B}{x - 1}$ (b) $\frac{2}{x^2 + x} = \frac{A}{x} + \frac{B}{x + 1}$
 3. $x + 6 \ln|x - 6| + C$ 5. $2 \ln|x + 5| - \ln|x - 2| + C$
 7. $\frac{1}{2} \ln \frac{3}{2}$ 9. $a \ln|x - b| + C$ 11. $2 \ln \frac{3}{2}$
 13. $\frac{27}{5} \ln 2 - \frac{9}{5} \ln 3$ (or $\frac{9}{5} \ln \frac{8}{3}$) 15. $2 + \ln \frac{25}{9}$
 17. $\ln[(e^x + 2)^2/(e^x + 1)] + C$
 19. $2 \ln|x| + 3 \ln|x + 2| + (1/x) + C$
 21. $A = 1, B = 1, C = 1; \ln|x| + \tan^{-1}x + \frac{1}{2} \ln|x^2 + 1| + C$

EXERCISES 5.7 | page 375

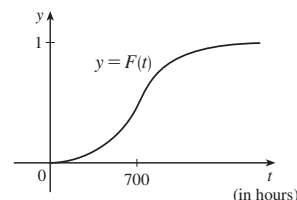
1. $\frac{1}{2\pi} \tan^2(\pi x) + \frac{1}{\pi} \ln|\cos(\pi x)| + C$
 3. $-\sqrt{4x^2 + 9}/(9x) + C$ 5. $\frac{1}{2}(e^{2x} + 1) \arctan(e^x) - \frac{1}{2}e^x + C$
 7. $\pi^3 - 6\pi$ 9. $-\frac{1}{2} \tan^2(1/z) - \ln|\cos(1/z)| + C$
 11. $\frac{1}{9} \sin^3 x [3 \ln(\sin x) - 1] + C$
 13. $\frac{1}{2\sqrt{3}} \ln \left| \frac{e^x + \sqrt{3}}{e^x - \sqrt{3}} \right| + C$ 15. $\frac{1}{5} \ln|x^5 + \sqrt{x^{10} - 2}| + C$
 17. $\frac{1}{2}(\ln x)\sqrt{4 + (\ln x)^2} + 2 \ln[\ln x + \sqrt{4 + (\ln x)^2}] + C$
 21. $\frac{1}{3} \tan x \sec^2 x + \frac{2}{3} \tan x + C$
 23. $\frac{1}{10}(1 + 2x)^{5/2} - \frac{1}{6}(1 + 2x)^{3/2} + C$
 25. $-\ln|\cos x| - \frac{1}{2} \tan^2 x + \frac{1}{4} \tan^4 x + C$
 27. $\frac{2}{5}(8 + 3\sqrt{x^2} - 4\sqrt{x})\sqrt{1 + \sqrt{x}} + C$

EXERCISES 5.8 | page 380

1. $\frac{1}{2} - 1/(2t^2)$; 0.495, 0.499995, 0.4999995; 0.5
 3. 2 5. Diverges 7. $2e^{-2}$ 9. Diverges 11. 0
 13. Diverges 15. $\frac{1}{25}$ 17. Diverges 19. $\pi/9$ 21. $\frac{1}{2}$
 23. e



25. (a) ≈ 4.23 $\mu\text{g/mL}$; 0.5 h
 (b) 5.75 ($\mu\text{g/mL}$) $\times$ hours; this is the long-term “availability” of a single drug dose.
 27. $\int_0^\infty P(I(x)) dx = \frac{a}{k}$ is the rate of photosynthesis per unit area in a water column of infinite depth. In practice, this is a good approximation for an entire water column of unit area in the deep ocean.
 29. (a)



- (b) The rate at which the fraction $F(t)$ increases as t increases
 (c) 1; all bulbs burn out eventually

33. $\sqrt{\frac{\pi}{2}}$ 35. $\frac{1}{2}\sqrt{\pi}$

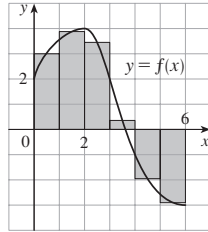
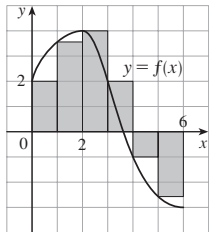
CHAPTER 5 REVIEW | page 381

True-False Quiz

1. True 3. True 5. False 7. True 9. True
11. False 13. False 15. False 17. False

Exercises

1. (a) 8 (b) 5.7

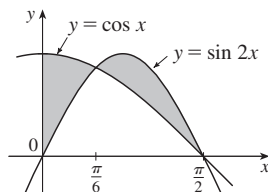


3. $\frac{1}{2} + \pi/4$ 5. 3 7. f is c , f' is b , $\int_0^x f(t) dt$ is a
9. 37 11. $\frac{9}{10}$ 13. $-(1/x) - 2 \ln|x| + x + C$ 15. $\frac{1}{2} \ln 2$
17. $\frac{1}{3} \sin 1$ 19. $(1/\pi)(e^\pi - 1)$ 21. $\sqrt{x^2 + 4x} + C$
23. $5 + 10 \ln \frac{2}{3}$ 25. 0 27. $\frac{64}{5} \ln 4 - \frac{124}{25}$
29. $3e^{\sqrt{x}}(x^{2/3} - 2x^{1/3} + 2) + C$ 31. $\ln|1 + \sec \theta| + C$
33. $\frac{64}{5}$ 35. $\frac{\cos^3 x}{1 + \sin^4 x}$ 37. $\frac{1}{2}[e^x \sqrt{1 - e^{2x}} + \sin^{-1}(e^x)] + C$
39. $4 \leq \int_1^3 \sqrt{x^2 + 3} dx \leq 4\sqrt{3}$ 41. $\frac{1}{36}$ 43. Diverges
45. The number of barrels of oil consumed from Jan. 1, 2000, through Jan. 1, 2015
47. 750 gallons
49. (a) $\int_a^b r(t) dt$
(b) $\int_a^b r(t) dt = (\text{number of births}) - (\text{number of deaths})$
51. $k - k \left(\frac{be^{-a} \sin b - ae^{-a} \cos b + a}{a^2 + b^2} \right)$
53. $e^{2x}(1 + 2x)/(1 - e^{-x})$

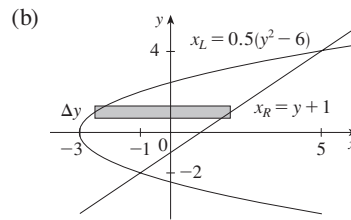
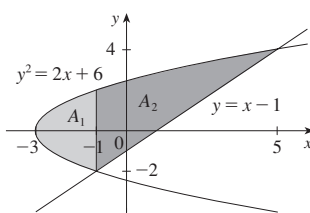
CHAPTER 6

EXERCISES 6.1 | page 392

1. $\frac{32}{3}$ 3. $e - (1/e) + \frac{4}{3}$ 5. $\frac{1}{3}$ 7. 72 9. $e - 2$
11. 0, 0.90; 0.04
13. $\frac{1}{2}$



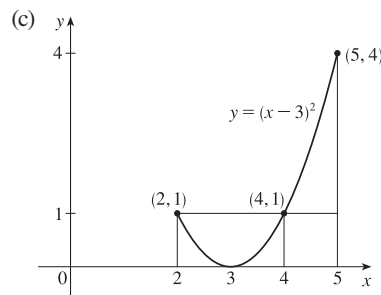
15. (a)



- (b) 18
17. 13 cm^2 19. 4232 cm^2
21. (a) $M_5 = 0.09 \text{ (mL/mL)} \cdot \text{min}$ (b) 711 mL/min
23. $117\frac{1}{3} \text{ ft}$
25. 8868; increase in population over a 10-year period
27. ± 6

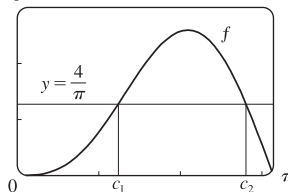
EXERCISES 6.2 | page 399

1. $\frac{8}{3}$ 3. $\frac{45}{28}$ 5. $2/(5\pi)$
7. (a) 1 (b) 2, 4



9. (a) $4/\pi$ (b) $\approx 1.24, 2.81$

- (c) 3

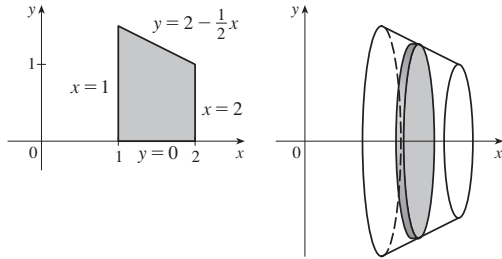
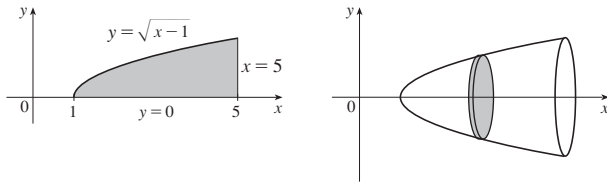
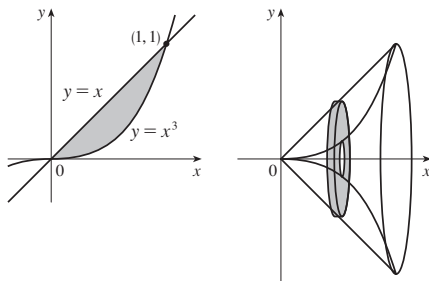
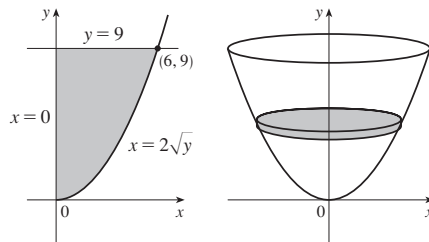


11. $\frac{9}{8}$ 13. $(50 + 28/\pi)^\circ\text{F} \approx 59^\circ\text{F}$
15. 74,719 million people
17. 845.25 cells/mL
19. $1/\pi \approx 0.32 \text{ L/s}$

EXERCISES 6.3 | page 404

1. (a) 1480 members (b) 9440 members
(c) Not all of the 9440 new members survive.
3. 21,046 insects 5. 48.3 mg 7. 12,417 gallons
9. $1.19 \times 10^{-4} \text{ cm}^3/\text{s}$ 11. 6.59 L/min 13. 5.69 L/min

■ EXERCISES 6.4 | page 411

1. $19\pi/12$ 3. 8π 5. $4\pi/21$ 7. 162π 9. $V \approx M_6 = 106.8 \text{ cm}^3$ 11. (a) 196 (b) 83813. $\frac{1}{3}\pi r^2 h$ 15. 24 17. $\frac{5}{12}\pi r^3$

■ CHAPTER 6 REVIEW | page 412

Exercises

1. $\frac{8}{3}$ 3. $\frac{7}{12}$ 5. $M_5 = 180.6 \text{ cm}^2$ 7. ≈ 0.007 9. $\approx 0.848 \text{ } \mu\text{g/mL}$ 11. 54,91613. 0.070 L/s

15. (a) 0.38 (b) 0.87

17. (a) $2\pi/15$ (b) $\pi/6$

19. 36

CHAPTER 7

■ EXERCISES 7.1 | page 428

1. Nonautonomous 3. Pure-time

5. (a) It must be either 0 or decreasing.

(c) $y = 0$ (d) $y = 1/(x+2)$ 7. (a) $0 < P < 4200$ (b) $P > 4200$ (c) $P = 0, P = 4200$

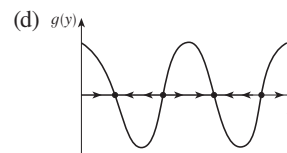
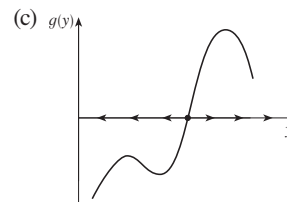
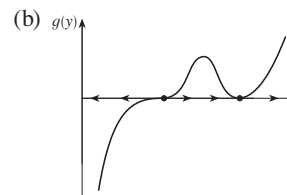
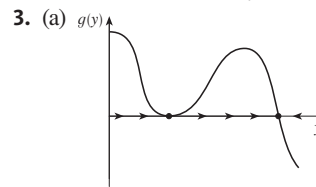
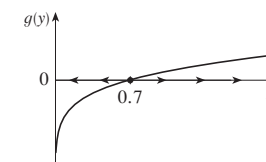
11. (a) III (b) I (c) IV (d) II

13. Pure-time; rate of change of drug concentration is a positive constant; $c(t) = kt$ 15. Nonautonomous; rate of change of concentration is proportional to the difference between the concentration and c_s , with a constant of proportionality k/t^p , which decreases over time

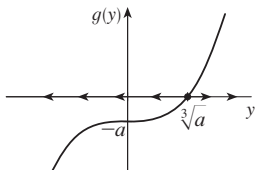
■ EXERCISES 7.2 | page 436

1. (a) (i) is locally stable.

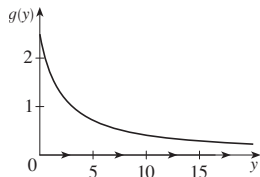
(b) (i) and (ii) are locally stable.

5. (a) $\hat{y} \approx 0.7035$ 

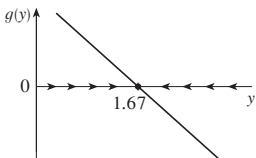
(b) $\hat{y} = \sqrt[3]{a}$



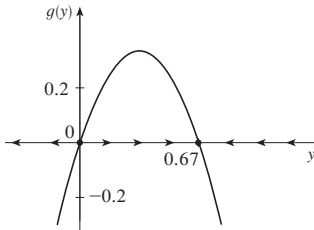
(c) No equilibria



7. (a) $\hat{y} = \frac{5}{3}$ is locally stable



(b) $\hat{y} = 0$ is unstable and $\hat{y} = \frac{2}{3}$ is locally stable



9. (a) $\hat{y} = -1/a$; locally stable when $a < 0$

(b) $\hat{y} = 0$; locally stable when $a < 0$

(c) $\hat{y} = \pi/2$; locally stable when $a > 0$

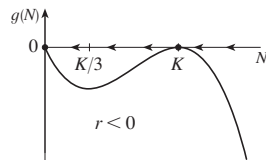
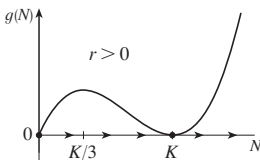
(d) $\hat{y} = 0$ is locally stable when $a < 0$;

$\hat{y} = a$ is locally stable when $a > 0$

11. (b) $r > 0$

(c) Slope of the phase plot at $N = K$ is zero (horizontal tangent line). Equilibrium may be locally stable or unstable.

(d) $\hat{N} = K$ is unstable in both cases.



13. (a) $\hat{N} = 0, \hat{N} = 500(2 - h)$ (b) $h \geq 2$ (c) $h < 2$

15. (a) $\hat{p} = 0, \hat{p} = 1 - (m/c)$

(b) $m < c$; colonization rate greater than extinction rate

(c) $m < c$

19. (a) $dp/dt = sp(1 - p) - \mu p$, where $s = r_1 - r_2$

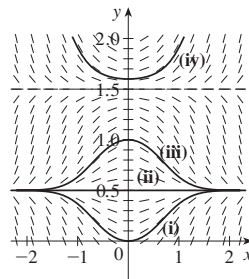
(b) $\hat{p} = 0, \hat{p} = 1 - \frac{\mu}{s}$

(c) $\hat{p} = 0$ is locally stable when $s < \mu$;

$\hat{p} = 1 - \frac{\mu}{s}$ is locally stable when $s > \mu$.

EXERCISES 7.3 | page 447

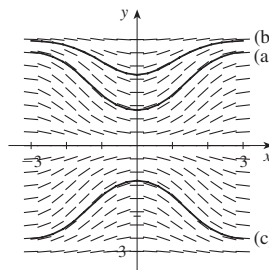
1. (a)



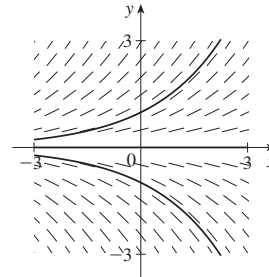
(b) $y = 0.5, y = 1.5$

3. III 5. IV

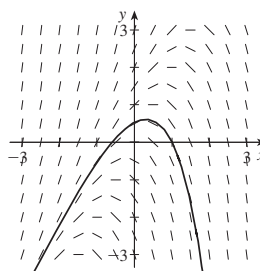
7.



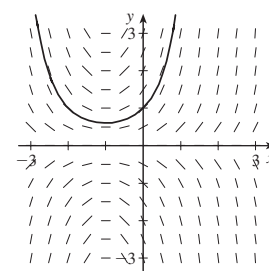
9.



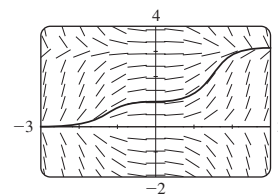
11.



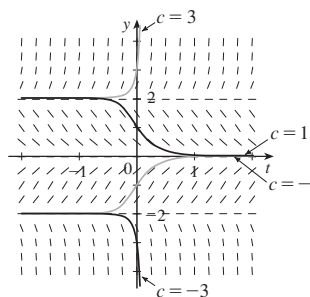
13.



15.

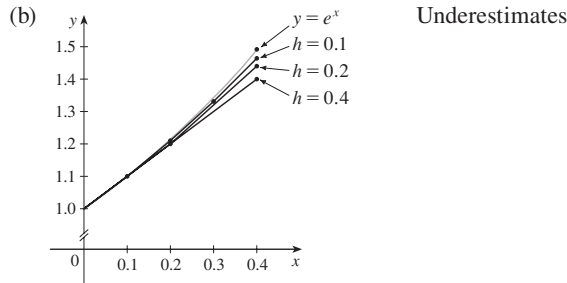


17.



$-2 \leq c \leq 2; -2, 0, 2$

19. (a) (i) 1.4 (ii) 1.44 (iii) 1.4641



- (c) (i) 0.0918 (ii) 0.0518 (iii) 0.0277

It appears that the error is also halved (approximately).

- 21.
- $N(4) \approx 1.3906$
- ; close agreement with Table 7.1.1

- 23.
- $-1, -3, -6.5, -12.25$
25. 1.7616

27. (a) (i) 3 (ii) 2.3928 (iii) 2.3701 (iv) 2.3681

- (c) (i)
- -0.6321
- (ii)
- -0.0249
- (iii)
- -0.0022
- (iv)
- -0.0002

It appears that the error is also divided by 10 (approximately).

EXERCISES 7.4 | page 455

1. $y = \frac{2}{K - x^2}, y = 0$ 3. $y = K\sqrt{x^2 + 1}$

5. $\frac{1}{2}y^2 - \cos y = \frac{1}{2}x^2 + \frac{1}{4}x^4 + C$

7. $y = \pm \sqrt{[3(te^t - e^t + C)]^{2/3} - 1}$

9. $u = Ae^{2t+t^2/2} - 1$

11. $y = -\sqrt{x^2 + 9}$ 13. $u = -\sqrt{t^2 + \tan t + 25}$

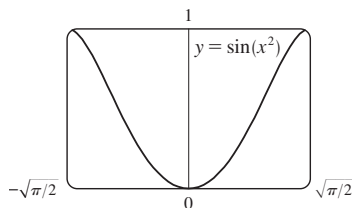
15. $\frac{1}{2}y^2 + \frac{1}{3}(3 + y^2)^{3/2} = \frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + \frac{41}{12}$

17. $y = \frac{4a}{\sqrt{3}} \sin x - a$

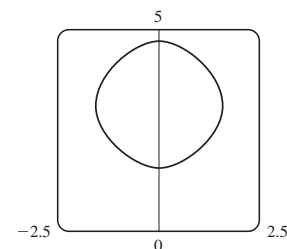
19. $y = e^{x^2/2}$ 21. $y = Ke^x - x - 1$

23. (a) $\sin^{-1}y = x^2 + C$ for $-\pi/2 \leq x^2 + C \leq \pi/2$; $y = 1$ is also a solution

(b) $y = \sin(x^2), -\sqrt{\pi/2} \leq x \leq \sqrt{\pi/2}$ (c) No

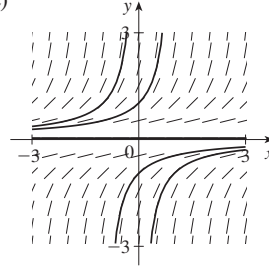


25. $\cos y = \cos x - 1$



27. (a), (c)

(b) $y = \frac{1}{K - x}, y = 0$



29. $y = 1 + e^{2-x^2/2}$ 31. $y = (\frac{1}{2}x^2 + 2)^2$

33. $n(t) = n_0 e^{1-(e^{-t}+t)}$; population goes extinct

35. $c(t) = c_s - c_s e^{-kt}$ 37. $n(t) = \frac{1}{4}(kt + 2)^2$

39. (a) $M(t) = \mu^{-3}[1 - (1 - \mu)e^{-\mu t/3}]^3$

(b) Approaches μ^{-3} grams

(c) $\frac{dD}{dt} = \frac{\mu}{3} \left(\frac{ka}{\mu} - D \right)$

41. $p(t) = \frac{p_0 e^{st}}{1 - p_0 + p_0 e^{st}}$

43. (a) $C(t) = (C_0 - r/k)e^{-kt} + r/k$

(b) r/k ; the concentration approaches r/k regardless of the value of C_0

45. (a) $15e^{-t/100}$ kg (b) $15e^{-0.2} \approx 12.3$ kg

47. About 4.9% 49. g/k

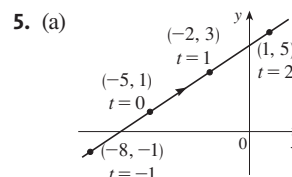
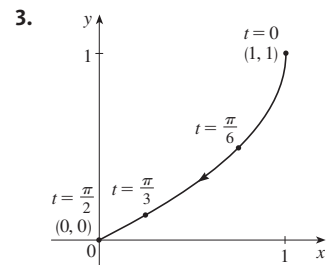
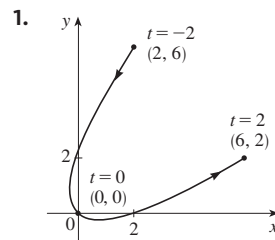
51. (a) $dA/dt = k\sqrt{A}(M - A)$

(b) $A(t) = M \left(\frac{Ce^{\sqrt{M}kt} - 1}{Ce^{\sqrt{M}kt} + 1} \right)^2$, where $C = \frac{\sqrt{M} + \sqrt{A_0}}{\sqrt{M} - \sqrt{A_0}}$

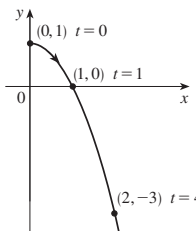
and $A_0 = A(0); A(t) = 0$

53. $S(A) = CA^k$, where S is number of species, A is the area, and C and k are constants

EXERCISES 7.5 | page 465



(b) $y = \frac{2}{3}x + \frac{13}{3}$

7. (a)  (b) $y = 1 - x^2, x \geq 0$

9. Moves counterclockwise along the circle $(x - 3)^2 + (y - 1)^2 = 4$ from $(3, 3)$ to $(3, -1)$

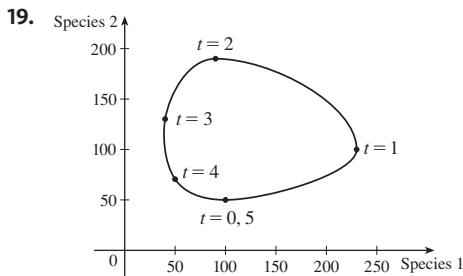
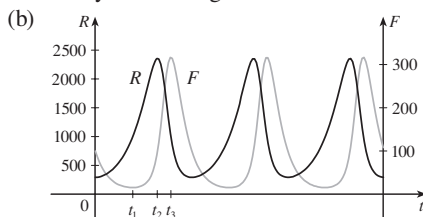
11. Moves 3 times clockwise around the ellipse $(x^2/25) + (y^2/4) = 1$, starting and ending at $(0, -2)$

13. (a) x = predators, y = prey; growth is restricted only by predators, which feed only on prey.

(b) x = prey, y = predators; growth is restricted by carrying capacity and by predators, which feed only on prey.

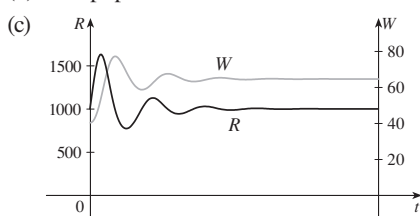
15. Competition

17. (a) The rabbit population starts at about 300, increases to 2400, then decreases back to 300. The fox population starts at 100, decreases to about 20, increases to about 315, decreases to 100, and the cycle starts again.

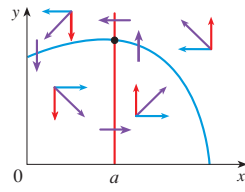


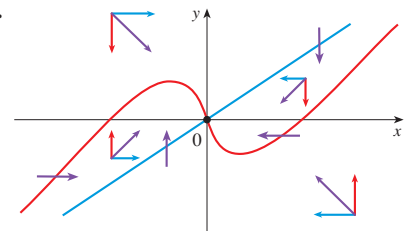
23. (a) Population stabilizes at 5000.

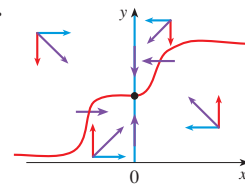
(b) The populations stabilize at 1000 rabbits and 64 wolves.

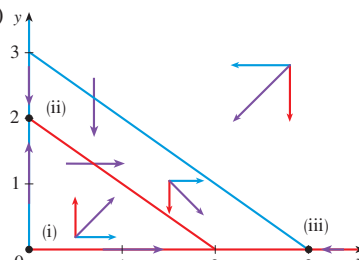


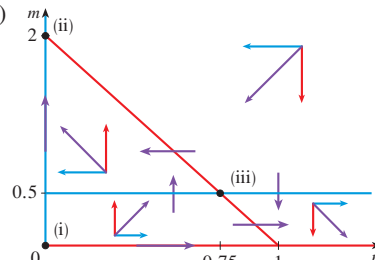
EXERCISES 7.6 | page 476

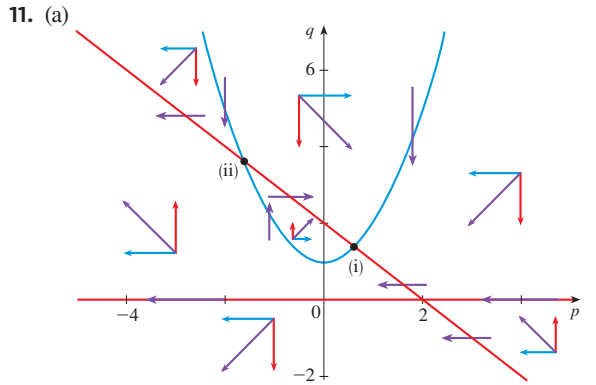
1.  Cannot tell

3.  Cannot tell

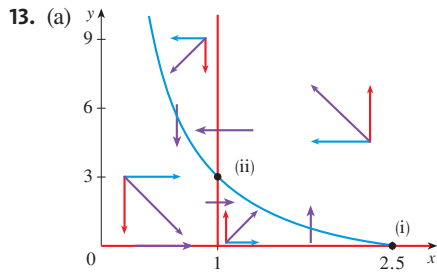
5.  Locally stable

7. (a) 
 (b) (i) $\hat{x} = 0, \hat{y} = 0$ (ii) $\hat{x} = 0, \hat{y} = 2$ (iii) $\hat{x} = 3, \hat{y} = 0$

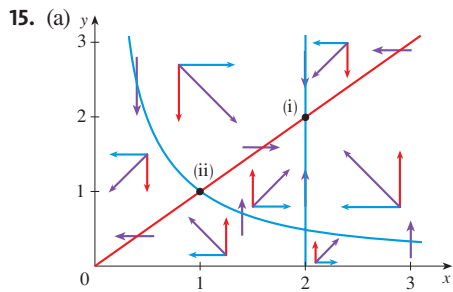
9. (a) 
 (b) (i) $\hat{n} = 0, \hat{m} = 0$ (ii) $\hat{n} = 0, \hat{m} = 2$ (iii) $\hat{n} = \frac{3}{4}, \hat{m} = \frac{1}{2}$



(b) (i) $\hat{p} \approx 0.618, \hat{q} \approx 1.382$ (ii) $\hat{p} \approx -1.618, \hat{q} \approx 3.618$



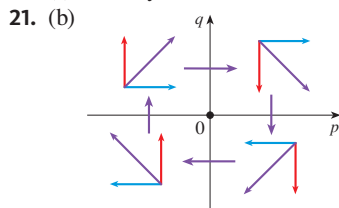
(b) (i) $\hat{x} = 2.5, \hat{y} = 0$ (ii) $\hat{x} = 1, \hat{y} = 3$



(b) (i) $\hat{x} = 2, \hat{y} = 2$ (ii) $\hat{x} = 1, \hat{y} = 1$

17. (a) Locally stable when $a > 1$ (b) $\hat{x} = 0, \hat{y} = 0$

19. (a) Locally stable for all values of a (b) $\hat{x} = a + 1, \hat{y} = 1$



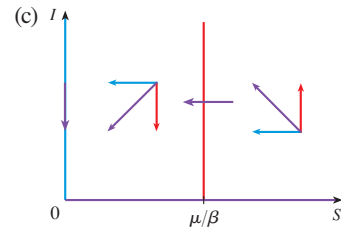
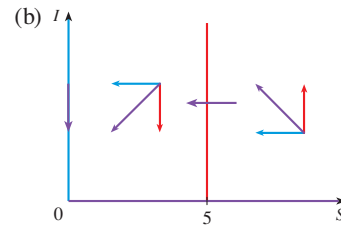
(c) The mass will cycle in front of and behind its rest position, $p = 0$, as time passes.

(d) The velocity of the mass will cycle from positive to negative values and back as time passes.

23. (a) $-\beta SI$: rate of decrease of the number of susceptible people due to transmission of the disease

βSI : rate of increase of the number of infected people due to transmission of the disease

$-\mu I$: rate of decrease of the infected population due to recovery

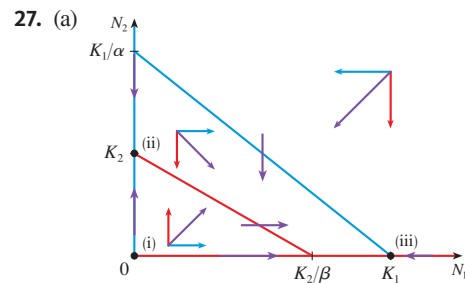
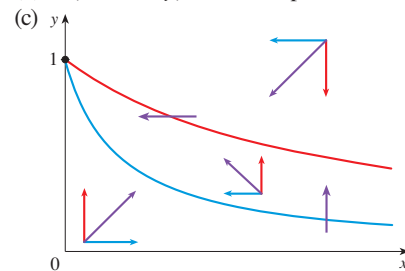


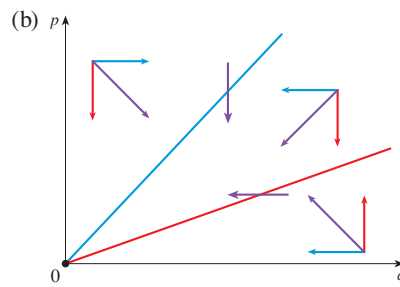
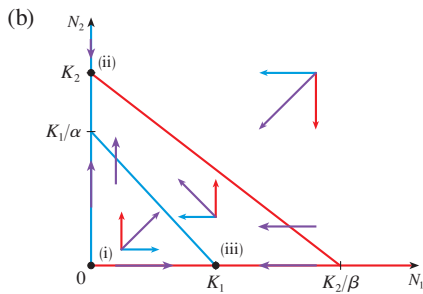
25. (a) $-k_f xyM$: rate of decrease of substrate molecules due to binding with enzymes, where k_f is the fractional rate of enzyme binding

$k_r(1 - y)M$: rate of increase of substrate molecules due to dissociation of the enzyme-substrate complex, where k_r is the fractional rate of dissociation

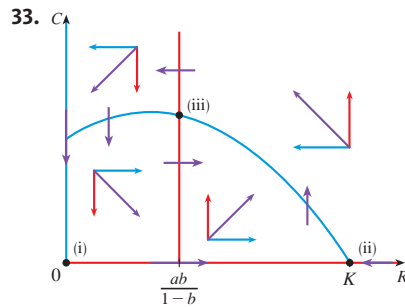
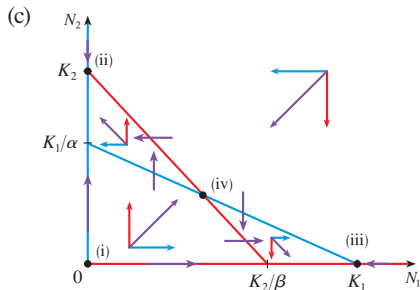
$k_{cat}(1 - y)M$: rate of increase of free enzymes due to the catalyzed reaction and rate at which product molecules are generated, where k_{cat} is the fractional rate of the forward reaction

(b) dx/dt and dy/dt are independent of z .





Urea concentration approaches zero in blood and pool.



CHAPTER 7 REVIEW | page 481

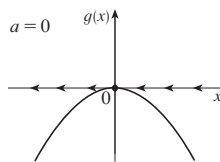
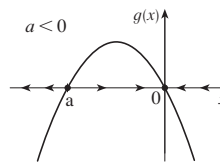
True-False Quiz

1. True 3. True 5. True

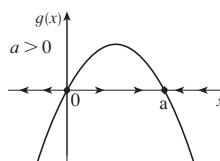
Exercises

1. (a) $\hat{x} = 0, a$

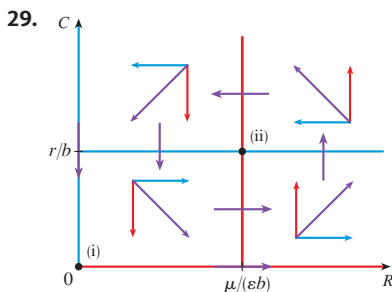
- (b) $a < 0$ (i) $\hat{x} = 0$ is stable,
 $\hat{x} = a$ is unstable



- (ii) $\hat{x} = a = 0$ is unstable



- (iii) $\hat{x} = 0$ is unstable,
 $\hat{x} = a$ is stable

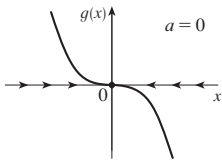
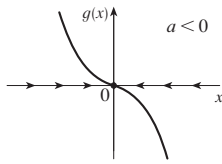


Oscillatory dynamics are predicted.

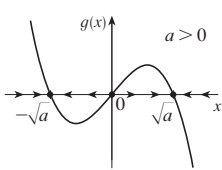
31. (a) $-\frac{K}{V}c$: rate of decrease of urea concentration in the blood due to dialysis, where K is the rate of flow through the dialyzer
 ap : rate of increase of urea concentration in the blood due to urea outflow from the inaccessible pool
 bc : rate of change of urea concentration due to urea flow from the blood back to the inaccessible pool

3. (a) $\hat{x} = 0, \pm\sqrt{a}$ (provided $a \geq 0$)

(b) $a < 0$ (i) $\hat{x} = 0$ is stable

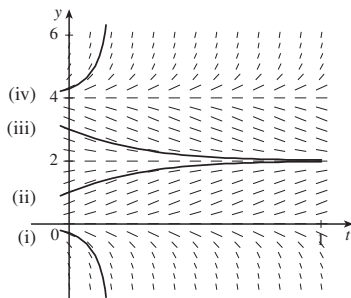


(ii) $\hat{x} = 0$ is stable



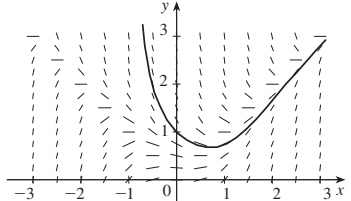
(iii) $\hat{x} = \pm\sqrt{a}$ are stable,
 $\hat{x} = 0$ is unstable

5. (a)



(b) $0 \leq c \leq 4$; $y = 0$, $y = 2$, $y = 4$ (c) Inconclusive

7. (a) $y(0.3) \approx 0.8$



(b) 0.75676

(c) $y = x$ and $y = -x$; there is a loc max or min

9. $y = \pm\sqrt{\ln(x^2 + 2x^{3/2} + C)}$

11. $r(t) = 5e^{t-t^2}$

13. $n(t) = n_0 e^{(365/2\pi) \sin(2\pi t/365)}$

15. (a) $p(t) = \frac{p_0(c-m)e^{(c-m)t}}{c-m-cp_0+p_0ce^{(c-m)t}}$ when $c \neq m$;

$p(t) = \frac{p_0}{cp_0t+1}$ when $c = m$

(b) $c \leq m$

17. (a) $V(t) = 3000 - 2400e^{-(1/300)t}$

(b) Approximately 416 s or 6.9 h

19. $k \ln h + h = (-R/V)t + C$

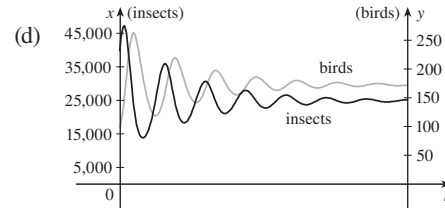
21. (a) Stabilizes at 200,000

(b) (i) $x = 0, y = 0$: zero populations

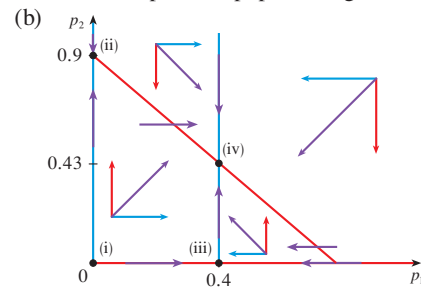
(ii) $x = 200,000, y = 0$: In the absence of birds, the insect population is always 200,000.

(iii) $x = 25,000, y = 175$: Both populations are stable.

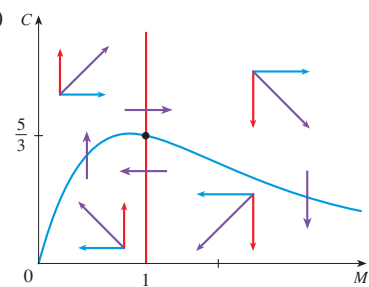
(c) The populations stabilize at 25,000 insects and 175 birds.



23. dp_1/dt has no term with p_2 so species 1 growth is unaffected by species 2; $-c_1p_1p_2$ term in dp_2/dt means species 2 population decreases as species 1 population grows.



25. (a)



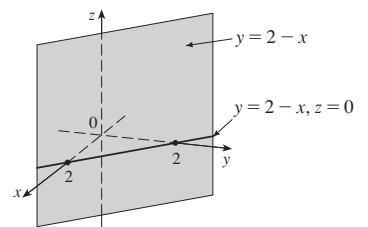
(b) M cycles between values above and below 1 mg/mL; cell division is a periodic process.

CHAPTER 8

EXERCISES 8.1 | page 493

1. $(4, 0, -3)$ 3. $Q; R$

5. A vertical plane that intersects the xy -plane in the line $y = 2 - x, z = 0$ (see graph at right)



7. (a) $|PQ| = 6$, $|QR| = 2\sqrt{10}$, $|RP| = 6$; isosceles triangle
 (b) $|PQ| = 3$, $|QR| = 3\sqrt{5}$, $|RP| = 6$; right triangle

9. (a) No (b) Yes

11. $(x - 3)^2 + (y - 8)^2 + (z - 1)^2 = 30$

13. $(3, -2, 1)$, 5 15. $(2, 0, -6)$, $9/\sqrt{2}$

17. (b) $\frac{5}{2}$, $\frac{1}{2}\sqrt{94}$, $\frac{1}{2}\sqrt{85}$

19. (a) $(x - 2)^2 + (y + 3)^2 + (z - 6)^2 = 36$

(b) $(x - 2)^2 + (y + 3)^2 + (z - 6)^2 = 4$

(c) $(x - 2)^2 + (y + 3)^2 + (z - 6)^2 = 9$

21. A plane parallel to the yz -plane and 5 units in front of it

23. A half-space consisting of all points to the left of the plane $y = 8$

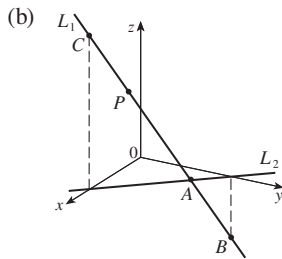
25. All points on or between the horizontal planes $z = 0$ and $z = 6$

27. All points on a circle with radius 2 and center on the z -axis that is contained in the plane $z = -1$

29. All points on or inside a sphere with radius $\sqrt{3}$ and center O

31. $0 < x < 5$ 33. $r^2 < x^2 + y^2 + z^2 < R^2$

35. (a) $(2, 1, 4)$



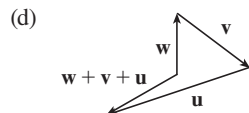
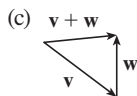
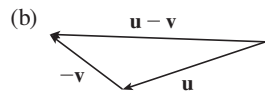
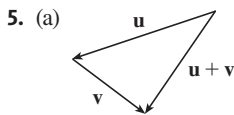
37. (a) 0.2 cm (b) 1.0 cm (c) ≈ 2.24 cm

39. (a) Yes (b) No (c) The circles in part (a) are the projections of the spheres in part (b) onto the xy -plane.

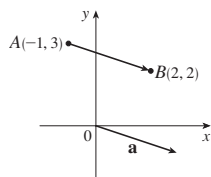
■ EXERCISES 8.2 | page 502

1. (a) Scalar (b) Vector (c) Vector (d) Scalar

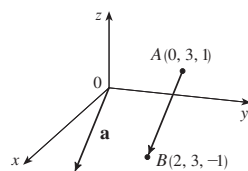
3. $\vec{AB} = \vec{DC}$, $\vec{DA} = \vec{CB}$, $\vec{DE} = \vec{EB}$, $\vec{EA} = \vec{CE}$



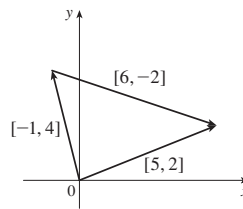
7. $\mathbf{a} = [3, -1]$



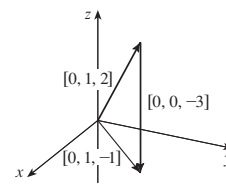
9. $\mathbf{a} = [2, 0, -2]$



11. $[5, 2]$



13. $[0, 1, -1]$



15. $[2, -18]$, $[1, -42]$, 13, 10

17. $[-1, 1, 2]$, $[-4, 1, 9]$, $\sqrt{14}$, $\sqrt{82}$

19. $\left[-\frac{3}{\sqrt{58}}, \frac{7}{\sqrt{58}}\right]$ 21. $\left[\frac{8}{9}, -\frac{1}{9}, \frac{4}{9}\right]$

23. $[2, 2\sqrt{3}]$ 25. ≈ 45.96 ft/s, ≈ 38.57 ft/s

27. $100\sqrt{7} \approx 264.6$ N, $\approx 139.1^\circ$

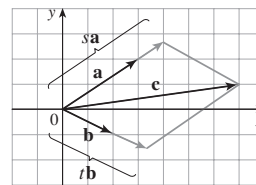
29. $\sqrt{493} \approx 22.2$ mi/h, $N8^\circ W$

31. (a) $5\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ (b) $-3\mathbf{i} + 2\mathbf{j} - 10\mathbf{k}$

(c) $14\mathbf{i} + 4\mathbf{j} + 15\mathbf{k}$ (d) $-23\mathbf{i} + 10\mathbf{j} - 64\mathbf{k}$

33. $\pm[1, 4]/\sqrt{17}$

35. (a), (b)



(c) $s \approx 1.3$, $t \approx 1.6$

(d) $s = \frac{9}{7}$, $t = \frac{11}{7}$

37. $\mathbf{a} \approx [0.50, 0.31, 0.81]$

39. (a) Observing a resultant voltage vector that points upwards and has a smaller (horizontal) magnitude compared to a healthy individual

- (b) Observing a resultant voltage vector that points to the right of the patient

41. A sphere with radius 1, centered at (x_0, y_0, z_0)

■ EXERCISES 8.3 | page 511

1. (b), (c), (d) are meaningful 3. -15 5. 14 7. 19

9. 1 11. $\mathbf{u} \cdot \mathbf{v} = \frac{1}{2}$, $\mathbf{u} \cdot \mathbf{w} = -\frac{1}{2}$

15. $\cos^{-1}\left(\frac{9 - 4\sqrt{7}}{20}\right) \approx 95^\circ$ 17. $\cos^{-1}\left(\frac{-1}{2\sqrt{7}}\right) \approx 101^\circ$

19. 45° , 45° , 90°

21. (a) Neither (b) Orthogonal (c) Orthogonal

- (d) Parallel

23. Yes 25. $\left[\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right]$ or $\left[-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$

27. $x - 2y + 5z = 0$

29. 3 , $\left[\frac{9}{5}, -\frac{12}{5}\right]$ 31. $1/\sqrt{21}$, $\left[\frac{2}{21}, -\frac{1}{21}, \frac{4}{21}\right]$

35. $[0, 0, -2\sqrt{10}]$ or any vector of the form $[s, t, 3s - 2\sqrt{10}]$, $s, t \in \mathbb{R}$

37. (a) Magnitude is about twice as large in North American viruses (b) Directions are perpendicular

39. (a) $\sqrt{115} \approx 10.7$

(b) Drug A: $37/115 \approx 0.32$ Drug B: $46/115 = 0.4$

(c) Drug B

41. $\cos^{-1}(1/\sqrt{3}) \approx 55^\circ$

43. Center: $[\frac{1}{2}(a_1 + b_1), \frac{1}{2}(a_2 + b_2), \frac{1}{2}(a_3 + b_3)]$

Radius: $\frac{1}{2}\sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$

■ EXERCISES 8.4 | page 518

1. (a) 2×2 (b) Not defined (c) 2×3 (d) 3×3
(e) Not defined (f) 3×2 (g) Not defined (h) 3×3

3. (a) $\begin{bmatrix} -25 & -1 \\ -20 & -23 \end{bmatrix}$

(b) $\begin{bmatrix} 3x + 13 & 5 & 26 \\ 16 & 3y + 3 & 51 \\ -2 & -10 & -4 \end{bmatrix}$

(c) $\begin{bmatrix} 32 & 9 & 68 \\ 135 & 5y & 23 \end{bmatrix}$ (d) Not defined

(e) $\begin{bmatrix} 33 & 5x - 5 \\ 5a - 1 & 18 \end{bmatrix}$ (f) Not defined

(g) $\begin{bmatrix} 2x & 4 & 18 \\ 12 & 2y & 26 \\ 0 & 2 & 0 \end{bmatrix}$ (h) $\begin{bmatrix} 8 & 3 & 20 \\ 45 & -y & 5 \end{bmatrix}$

5. (a) $[3X \ 1 \ 2]$ (b) $\begin{bmatrix} 3 \\ 3 \\ 9 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 8 \\ 1 & 3 \\ 7 & 6 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

7. (b) d_{ii}^k

9. (a) $C^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, C^3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, C^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, C^5 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b)

$$C^k = \begin{cases} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} & \text{when } k \text{ is odd} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \text{when } k \text{ is even} \end{cases}$$

17. $a_{ij} = \begin{cases} 0 & \text{when } i = j \text{ (diagonal terms)} \\ -a_{ji} & \text{when } i \neq j \text{ (off-diagonal terms)} \end{cases}$

■ EXERCISES 8.5 | page 525

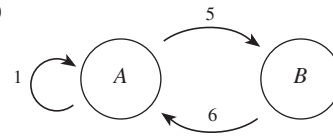
1. (a) $\begin{bmatrix} X & 0 \\ X & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & X & 0 \\ X & 0 & X \\ 0 & X & 0 \end{bmatrix}$ (c) $\begin{bmatrix} X & X & 0 & 0 \\ X & 0 & X & 0 \\ 0 & X & X & X \\ 0 & 0 & X & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & X & 0 \\ X & 0 & 0 \\ 0 & 0 & X \end{bmatrix}$ (e) $\begin{bmatrix} X & X \\ X & X \end{bmatrix}$

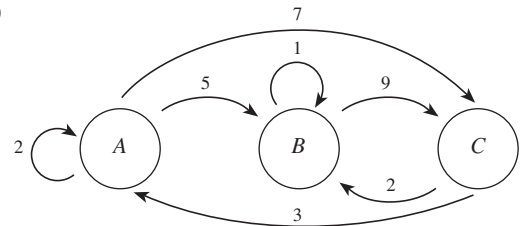
3. (a) $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 & 4 \\ 1 & 2 & 0 \\ 0 & 3 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$

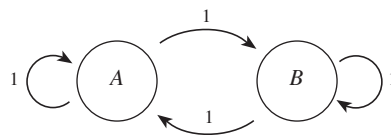
5. (a)



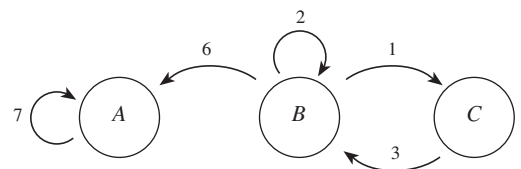
(b)



(c)

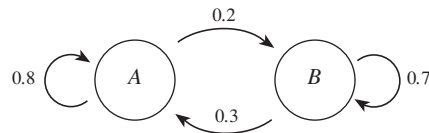


(d)

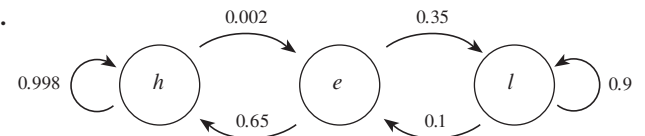


7. No contributions to the variable having a row of zeroes; a matrix diagram would have no incoming arrows to the circle representing this variable.

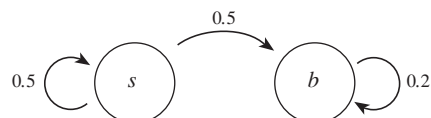
9.



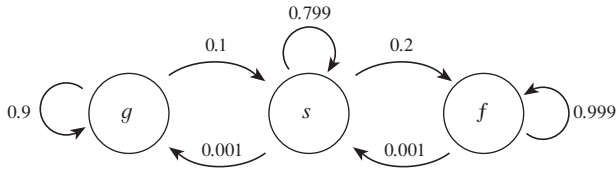
11.



13.



15.



$$17. \begin{cases} x_{t+1} = 2x_t + 3y_t \\ y_{t+1} = 0.9y_t \end{cases}, \begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 0.9 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix}$$

$$19. \begin{cases} d_{t+1} = 1d_t + \frac{1}{4}h_t \\ h_{t+1} = \frac{1}{2}h_t \\ r_{t+1} = \frac{1}{4}h_t + 1r_t \end{cases}, \begin{bmatrix} d_{t+1} \\ h_{t+1} \\ r_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & 1 \end{bmatrix} \begin{bmatrix} d_t \\ h_t \\ r_t \end{bmatrix}$$

$$21. \begin{cases} X_{t+1} = 0.92X_t + 0.9Z_t \\ Y_{t+1} = 0.05X_t + 0.999Y_t \\ Z_{t+1} = 0.03X_t + 0.001Y_t + 0.1Z_t \end{cases},$$

$$\begin{bmatrix} X_{t+1} \\ Y_{t+1} \\ Z_{t+1} \end{bmatrix} = \begin{bmatrix} 0.92 & 0 & 0.9 \\ 0.05 & 0.999 & 0 \\ 0.03 & 0.001 & 0.1 \end{bmatrix} \begin{bmatrix} X_t \\ Y_t \\ Z_t \end{bmatrix}$$

EXERCISES 8.6 | page 534

1. (a) Yes (b) Yes (c) No (d) Yes (e) No (f) Yes

$$5. \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix}$$

11. 1 13. 1, 2 15. $\det A^k \neq 0$, so A^k is nonsingular.19. One solution when $k \neq 6$; infinite number of solutions when $k = 6$

$$21. x_1 = \frac{a_{22}b_1 - a_{12}b_2}{a_{11}a_{22} - a_{12}a_{21}}, x_2 = \frac{-a_{21}b_1 + a_{11}b_2}{a_{11}a_{22} - a_{12}a_{21}}$$

$$25. x_1 = \frac{11}{4}, x_2 = -\frac{3}{4} \quad 27. x_1 = -1, x_2 = \frac{5}{6}$$

29. Infinite number of solutions satisfying $x_2 = -2x_1 + 1$

$$31. \mathbf{x}_t = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$33. \hat{\mathbf{n}} = \begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix} \hat{\mathbf{n}} \Rightarrow \hat{\mathbf{n}} - \begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix} \hat{\mathbf{n}} = \mathbf{0}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \hat{\mathbf{n}} - \begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix} \hat{\mathbf{n}} = \mathbf{0}$$

$$\Rightarrow \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix} \right) \hat{\mathbf{n}} = \mathbf{0}$$

$$\Rightarrow \underbrace{\begin{bmatrix} 1-b & -2 \\ -\frac{1}{2} & 1 \end{bmatrix}}_A \hat{\mathbf{n}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So finding $\hat{\mathbf{n}}$ requires solving the homogeneous system above. The matrix A will be invertible when $\det A \neq 0 \Rightarrow$

$$(1-b)(1) - (-\frac{1}{2})(-2) \neq 0 \Rightarrow b \neq 0.$$

$$(a) \hat{\mathbf{n}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(b) $\hat{\mathbf{n}} = \begin{bmatrix} 2k \\ k \end{bmatrix}$, where k is any real number (infinite number of equilibria)35. $\hat{\mathbf{v}} = \begin{bmatrix} k \\ 0 \end{bmatrix}$, where k is any real number (infinite number of equilibria)

EXERCISES 8.7 | page 544

1. (a) Reflects about the x -axis (b) Reflects about the y -axis(c) Reflects about the x -axis and the y -axis(d) Reflects about the x - and y -axes and the line $y = x$ (swaps x - and y -values).3. Vertical segment of L rotates clockwise and increases in length(b) Horizontal segment of L rotates clockwise and increases in length(c) Vertical segment of L rotates counterclockwise and increases in length(d) Horizontal segment of L rotates counterclockwise and increases in length

$$5. (a) \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & 1.5 \\ 0 & 1 \end{bmatrix}$$

$$(d) \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

7. Denote the voltage vector of a healthy heart by $\mathbf{v} = [0.3, -0.2]$.

$$(a) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (b) \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad (c) \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$(d) \begin{bmatrix} 1 & 1 \\ 0 & 0.3335 \end{bmatrix}$$

9. (a) Yes (b) Yes (c) No (d) Yes (e) No (f) Yes

11. (a) 0, 2 (b) 1, -1 (c) 2, 3 (d) $i, -i$ (e) 0, 2

(f) 1, 2, 3

$$13. (a) \lambda = 1, \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \lambda = -1, \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$(b) \lambda = 3, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \lambda = -1, \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$(c) \lambda = 1 + 2i, \mathbf{v} = \begin{bmatrix} i \\ 1 \end{bmatrix}; \lambda = 1 - 2i, \mathbf{v} = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$(d) \lambda = 2, \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \lambda = 5, \mathbf{v} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

$$(e) \lambda = -1 + \sqrt{10}, \mathbf{v} = \begin{bmatrix} -2 \\ 2 - \sqrt{10} \end{bmatrix};$$

$$\lambda = -1 - \sqrt{10}, \mathbf{v} = \begin{bmatrix} -2 \\ 2 + \sqrt{10} \end{bmatrix}$$

$$(f) \lambda = 1 + \sqrt{31}, \mathbf{v} = \begin{bmatrix} -6 \\ 1 - \sqrt{31} \end{bmatrix};$$

$$\lambda = 1 - \sqrt{31}, \mathbf{v} = \begin{bmatrix} -6 \\ 1 + \sqrt{31} \end{bmatrix}$$

$$(g) \lambda = 1, \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \lambda = 1 + \sqrt{3}, \mathbf{v} = \begin{bmatrix} \sqrt{3} \\ 2 \\ 3 \end{bmatrix};$$

$$\lambda = 1 - \sqrt{3}, \mathbf{v} = \begin{bmatrix} -\sqrt{3} \\ 2 \\ 3 \end{bmatrix}$$

$$(h) \lambda = 1, \mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \lambda = 1 + \sqrt{14}, \mathbf{v} = \begin{bmatrix} 3 + \sqrt{14} \\ 7 \\ \sqrt{14} \end{bmatrix};$$

$$\lambda = 1 - \sqrt{14}, \mathbf{v} = \begin{bmatrix} -3 + \sqrt{14} \\ -7 \\ \sqrt{14} \end{bmatrix}$$

15. (a) Free to choose any values for the components of the eigenvectors associated with the repeated eigenvalue $\lambda = 1$.

Two distinct eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are associated with $\lambda = 1$.

(b) Only one eigenvector $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is associated with the repeated eigenvalue $\lambda = 1$.

(c) Two distinct eigenvectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ are associated with

the repeated eigenvalue $\lambda = 1$. One eigenvector $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is associated with $\lambda = 2$.

25. (a) $\lambda_A + \lambda_B$ (b) $\lambda_A \lambda_B$

$$27. \lambda = 2, \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \lambda = \frac{9}{10}, \mathbf{v} = \begin{bmatrix} 30 \\ -11 \end{bmatrix}$$

$$29. \lambda = \frac{1}{2}b + \frac{1}{2}\sqrt{b^2 + 4}, \mathbf{v} = \begin{bmatrix} 4 \\ -b + \sqrt{b^2 + 4} \end{bmatrix};$$

$$\lambda = \frac{1}{2}b - \frac{1}{2}\sqrt{b^2 + 4}, \mathbf{v} = \begin{bmatrix} -4 \\ b + \sqrt{b^2 + 4} \end{bmatrix}$$

$$31. -\lambda^3 + \lambda^2 + \lambda + \frac{2}{3}$$

■ EXERCISES 8.8 | page 556

$$1. D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, P^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$3. D = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}, P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$5. D = \begin{bmatrix} 2 + \sqrt{5}i & 0 \\ 0 & 2 - \sqrt{5}i \end{bmatrix}, P = \begin{bmatrix} 2 & 2 \\ 1 + \sqrt{5}i & 1 - \sqrt{5}i \end{bmatrix},$$

$$P^{-1} = \begin{bmatrix} \frac{1}{4\sqrt{5}}i + \frac{1}{4} & -\frac{1}{2\sqrt{5}}i \\ -\frac{1}{4\sqrt{5}}i + \frac{1}{4} & \frac{1}{2\sqrt{5}}i \end{bmatrix}$$

$$7. D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, P^{-1} = \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$9. D^2 = \begin{bmatrix} 1 & 0 \\ 0 & 16 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, P^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$11. D^2 = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}, P = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}, P^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$$

$$13. D^2 = \begin{bmatrix} a^2 & 0 \\ 0 & b^2 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, P^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$15. \mathbf{n}_t = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} 2^t \quad 17. \mathbf{n}_t = \begin{bmatrix} 1 \\ 1 \end{bmatrix} 2^t$$

$$19. \mathbf{n}_t = \begin{bmatrix} 1 - a \\ 0 \end{bmatrix} + \begin{bmatrix} a \\ 1 \end{bmatrix} 2^t$$

21–25 Denote the arbitrary initial condition by $\mathbf{n}_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$.

$$21. \mathbf{n}_t = x_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} a^t + y_0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} b^t$$

$$23. \mathbf{n}_t = -y_0 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + (x_0 + y_0) \begin{bmatrix} 1 \\ 0 \end{bmatrix} 2^t$$

$$25. \mathbf{n}_t = \frac{1}{2}(x_0 + y_0) \begin{bmatrix} 1 \\ 1 \end{bmatrix} (a + b)^t + \frac{1}{2}(x_0 - y_0) \begin{bmatrix} 1 \\ -1 \end{bmatrix} (a - b)^t$$

$$31. \mathbf{n}_t = (\sqrt{2})^t \begin{bmatrix} \cos\left(\frac{\pi}{4}t\right) & -\sin\left(\frac{\pi}{4}t\right) \\ \sin\left(\frac{\pi}{4}t\right) & \cos\left(\frac{\pi}{4}t\right) \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$33. \mathbf{n}_t = (\sqrt{6})^t \begin{bmatrix} \sqrt{5} \sin(\theta t) \\ \cos(\theta t) \end{bmatrix}, \text{ where } \theta = \tan^{-1} \sqrt{5}$$

$$35. (a) \text{ No } (b) \mathbf{v}_t = \begin{bmatrix} 0.3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.2 \end{bmatrix} (-1)^t$$

$$(c) \mathbf{v}_t \text{ cycles between } \begin{bmatrix} 0.3 \\ -0.2 \end{bmatrix} \text{ and } \begin{bmatrix} 0.3 \\ 0.2 \end{bmatrix}.$$

$$37. (a) \lambda_1 = 1, \lambda_2 = \frac{7}{10}$$

(b) $\mathbf{y}_t$ approaches steady state in long term.

$$(c) \mathbf{y}_t = \frac{1}{3} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \left(\frac{7}{10} \right)^t \right)$$

$$(d) \lim_{t \rightarrow \infty} \mathbf{y}_t = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$39. (a) \lambda_1 = 1, \lambda_2 = \frac{3}{10}$$

(b) $\mathbf{x}_t$ approaches steady state in long term.

$$(c) \mathbf{x}_t = \frac{1}{7} \begin{bmatrix} 5 \\ 2 \end{bmatrix} + \left(a - \frac{5}{7} \right) \begin{bmatrix} 1 \\ -1 \end{bmatrix} \left(\frac{3}{10} \right)^t$$

$$(d) \lim_{t \rightarrow \infty} \mathbf{x}_t = \frac{1}{7} \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

41. (b) $\lambda_1 = \frac{1 + \sqrt{5}}{2}, \lambda_2 = \frac{1 - \sqrt{5}}{2}$

(c) Yes; components of $\mathbf{z}_n$ grow by a factor λ_1 each time step and asymptotically approach the line defined by the associated eigenvector $\mathbf{v}_1$.

(d) $F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$

CHAPTER 8 REVIEW | page 561

True-False Quiz

1. True 3. False 5. False 7. False 9. True
11. True 13. False 15. True 17. True

Exercises

1. (a) $(x + 1)^2 + (y - 2)^2 + (z - 1)^2 = 69$

(b) $(y - 2)^2 + (z - 1)^2 = 68, x = 0$

(c) Center $(4, -1, -3)$, radius 5

3. $(x_1 - c_1)^2 + (x_2 - c_2)^2 + \cdots + (x_n - c_n)^2 = r^2$ for a hypersphere centered at $P(c_1, c_2, \dots, c_n)$ with radius r

5. $-2, -4$ 7. (a) 45° (b) 74.7° 9. $\frac{13}{5}$

11. (a) $\sqrt{8} \approx 2.83$ (b) $\cos^{-1}\left(\frac{1}{\sqrt{5}}\right) \approx 1.11$ (or 63.4°)

13. $\begin{bmatrix} 0 & 0 & 0 \\ 0.5 & 1 & 2 \\ 0 & 0.5 & 0 \end{bmatrix}$

15. (a) Not defined (b) $\begin{bmatrix} 9 & 4 \\ 19 & 21 \end{bmatrix}$ (c) Not defined

(d) Not defined (e) $\begin{bmatrix} 16 & 1 & 7 \\ 4 & 5 & 11 \end{bmatrix}$ (f) $\begin{bmatrix} \frac{13}{7} & 4 \\ \frac{1135}{63} & \frac{107}{9} \end{bmatrix}$

21. $x = \frac{9}{11}, y = \frac{5}{11}$

23. Infinite number of solutions given by $y = -7x$

25. $\hat{\mathbf{n}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$; population extinct

27. $\lambda = 2, \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}; \lambda = 4, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

29. $\lambda = 2 + i, \mathbf{v} = \begin{bmatrix} 1 \\ i \end{bmatrix}; \lambda = 2 - i, \mathbf{v} = \begin{bmatrix} i \\ 1 \end{bmatrix}$

33. $\mathbf{n}_t = -\frac{1}{2} \begin{bmatrix} 1 \\ -2 \end{bmatrix} (-4)^t$ for $t \geq 1$

35. $\mathbf{n}_t = r^t \left(\begin{bmatrix} \cos(\theta t) - \sin(\theta t) \\ \sin(\theta t) + \cos(\theta t) \end{bmatrix} \right)$

where $r = \sqrt{1 + a^2}$ and $\theta = \tan^{-1}\left(\frac{1}{a}\right)$

37. Yes

39. (a) $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{bmatrix}$ (c) $\lambda_1 = 1$ and $\lambda_2 = -\frac{1}{2}$

(d) $\mathbf{z}_n$ approaches steady state in long term.

(e) $\lim_{n \rightarrow \infty} a_n = c$ (component 1 of $\mathbf{v}_1$), where c is constant

CHAPTER 9

EXERCISES 9.1 | page 580

1. (a) -27°C ; a temperature of -15°C with wind blowing at 40 km/h feels equivalent to about -27°C without wind.

(b) When the temperature is -20°C , what wind speed gives a wind chill of -30°C ? 20 km/h

(c) With a wind speed of 20 km/h, what temperature gives a wind chill of -49°C ? -35°C

(d) A function of wind speed that gives wind-chill values when the temperature is -5°C

(e) A function of temperature that gives wind-chill values when the wind speed is 50 km/h

3. (a) ≈ 20.5 ; the surface area of a person 70 in tall who weighs 160 lb is approximately 20.5 square feet.

(b) Answers vary depending on height and weight.

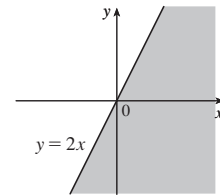
5. (a) ≈ 94.2 ; When the manufacturer invests \$20 million and the number of labor hours is 120,000, its yearly production is about \$94.2 million.

7. Snake with $R = 3$ and $S = 1$ is likely to survive longer.

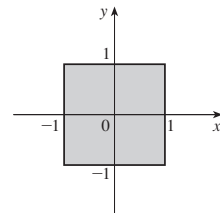
9. (a) 1 (b) $\mathbb{R}^2$ (c) $[-1, 1]$

11. (a) 3 (b) $\{(x, y, z) \mid x^2 + y^2 + z^2 < 4, x \geq 0, y \geq 0, z \geq 0\}$, interior of a sphere of radius 2, center the origin, in the first octant

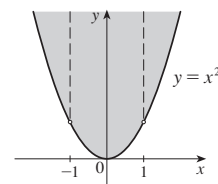
13. $\{(x, y) \mid y \leq 2x\}$



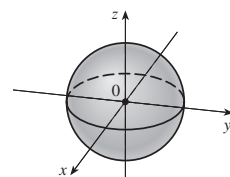
15. $\{(x, y) \mid -1 \leq x \leq 1, -1 \leq y \leq 1\}$



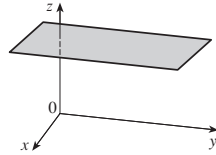
17. $\{(x, y) \mid y \geq x^2, x \neq \pm 1\}$



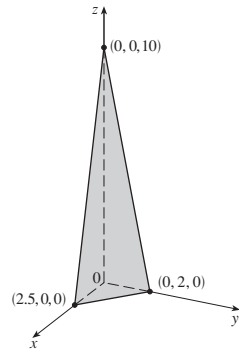
19. $\{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$



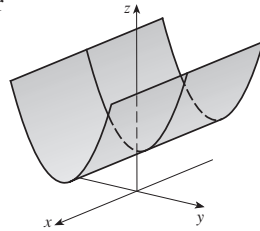
21. $z = 3$, horizontal plane



23. $4x + 5y + z = 10$, plane



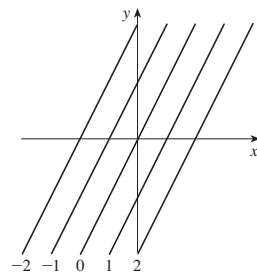
25. $z = y^2 + 1$, parabolic cylinder



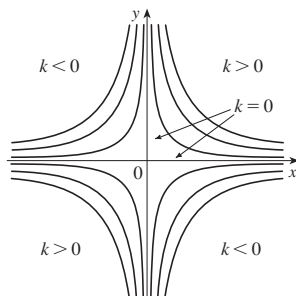
27. ≈ 56 , ≈ 35 29. 11°C , 19.5°C

31. Steep; nearly flat

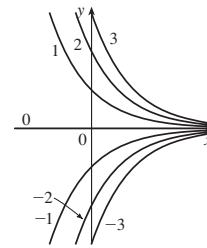
33. $y = 2x - k$



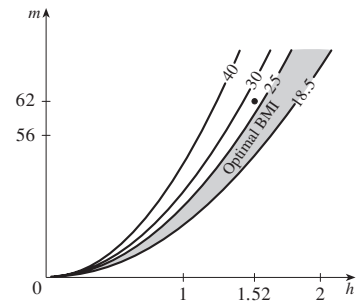
35. $y = k/x$



37. $y = ke^{-x}$



39. $m = kh^2$

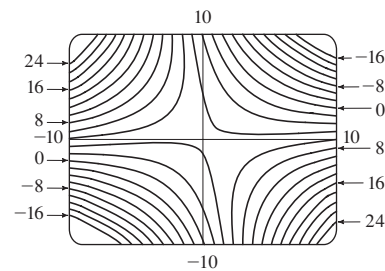
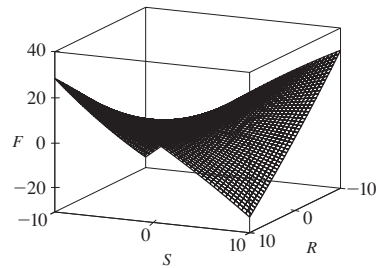


No

41. (a) C (b) II 43. (a) F (b) I

45. (a) B (b) VI

47.



$F(R, S)$ is shaped like a saddle.

49. 1 51. $\frac{2}{7}$ 53. Does not exist 55. Does not exist

57. 0 59. $\{(x, y) \mid y \geq 0\}$ 61. $\{(x, y) \mid x^2 + y^2 > 4\}$

63. $\{(x, y, z) \mid y \geq 0, y \neq \sqrt{x^2 + z^2}\}$

65. $\{(x, y) \mid (x, y) \neq (0, 0)\}$

EXERCISES 9.2 | page 593

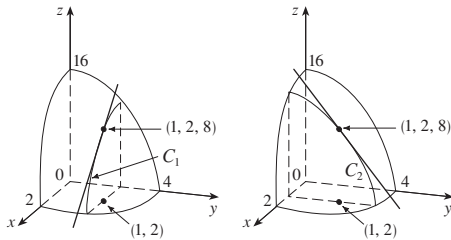
1. (a) The rate of change of temperature as longitude varies, with latitude and time fixed; the rate of change as only latitude varies; the rate of change as only time varies
- (b) Positive, negative, positive

3. (a) $f_T(-15, 30) \approx 1.3$; for a temperature of -15°C and wind speed of 30 km/h, the wind-chill index rises by 1.3°C for each degree the temperature increases. $f_v(-15, 30) \approx -0.15$; for a temperature of -15°C and wind speed of 30 km/h, the wind-chill index decreases by 0.15°C for each km/h the wind speed increases.

(b) Positive, negative (c) 0

5. (a) Positive (b) Negative

7. $f_x(1, 2) = -8 = \text{slope of } C_1$, $f_y(1, 2) = -4 = \text{slope of } C_2$



9. $f_x(x, y) = -3y$, $f_y(x, y) = 5y^4 - 3x$

11. $f_x(x, t) = -\pi e^{-t} \sin \pi x$, $f_t(x, t) = -e^{-t} \cos \pi x$

13. $\partial z/\partial x = 20(2x + 3y)^9$, $\partial z/\partial y = 30(2x + 3y)^9$

15. $f_x(x, y) = 2y/(x + y)^2$, $f_y(x, y) = -2x/(x + y)^2$

17. $\partial w/\partial \alpha = \cos \alpha \cos \beta$, $\partial w/\partial \beta = -\sin \alpha \sin \beta$

19. $f_r(r, s) = \frac{2r^2}{r^2 + s^2} + \ln(r^2 + s^2)$, $f_s(r, s) = \frac{2rs}{r^2 + s^2}$

21. $\partial u/\partial t = e^{w/t}(1 - w/t)$, $\partial u/\partial w = e^{w/t}$

23. $f_x = z - 10xy^3z^4$, $f_y = -15x^2y^2z^4$, $f_z = x - 20x^2y^3z^3$

25. $\partial w/\partial x = 1/(x + 2y + 3z)$, $\partial w/\partial y = 2/(x + 2y + 3z)$, $\partial w/\partial z = 3/(x + 2y + 3z)$

27. $\partial u/\partial x = e^{-t} \sin \theta$, $\partial u/\partial t = -xe^{-t} \sin \theta$, $\partial u/\partial \theta = xe^{-t} \cos \theta$

29. $f_x = yz^2 \tan(yt)$, $f_y = xyz^2 t \sec^2(yt) + xz^2 \tan(yt)$, $f_z = 2xyz \tan(yt)$, $f_t = xyz^2 t \sec^2(yt)$

31. $\partial u/\partial x_i = x_i/\sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$ 33. $\frac{1}{5}$ 35. $\frac{1}{4}$

37. $\frac{\partial z}{\partial x} = \frac{3yz - 2x}{2z - 3xy}$, $\frac{\partial z}{\partial y} = \frac{3xz - 2y}{2z - 3xy}$

39. $\frac{\partial z}{\partial x} = \frac{1 + y^2z^2}{1 + y + y^2z^2}$, $\frac{\partial z}{\partial y} = \frac{-z}{1 + y + y^2z^2}$

41. (a) $\approx 0.0545 \text{ ft}^2/\text{lb}$; rate at which body surface area increases with respect to weight when an individual weighs 160 lb and has a height 70 in.

(b) $\approx 0.213 \text{ ft}^2/\text{in}$; rate at which body surface area increases with respect to height when an individual weighs 160 lb and has a height 70 in.

43. $\frac{\partial R}{\partial L} = \frac{C}{r^4}$, rate of increase of resistance of blood when artery length increases (with constant artery radius);

$\frac{\partial R}{\partial r} = -4C \frac{L}{r^5}$, rate of change of resistance of blood when artery radius increases (with constant artery length)

45. $\frac{\partial P}{\partial v} = 3Av^2 - \frac{B(mg/x)^2}{v^2}$, rate of change in required power that occurs from an increase in the bird's velocity;

$\frac{\partial P}{\partial x} = -2 \frac{Bm^2g^2}{vx^3}$, rate of change in required power that occurs from an increase in the fraction of time the bird spends flapping;

$\frac{\partial P}{\partial m} = 2 \frac{Bmg^2}{vx^2}$, rate of change in required power that occurs from an increase in the bird's mass

47. $E_m(400, 8) \approx 0.301 \text{ kcal/g}$, an increase in mass of 1 g requires approximately 0.3 kcal of additional energy when $m = 400$ and $v = 8$.

$E_v(400, 8) \approx -4.89 \text{ kcal/(km/h)}$, an increase in speed of 1 km/h results in approximately 4.9 kcal of energy saved when $m = 400$ and $v = 8$.

49. $\frac{\partial T}{\partial P} = \frac{V - nb}{nR}$, $\frac{\partial P}{\partial V} = \frac{2n^2a}{V^3} - \frac{nRT}{(V - nb)^2}$

51. $f_{xx} = 6xy^5 + 24x^2y$, $f_{xy} = 15x^2y^4 + 8x^3 = f_{yx}$, $f_{yy} = 20x^3y^3$

53. $w_{uu} = v^2/(u^2 + v^2)^{3/2}$, $w_{uv} = -uv/(u^2 + v^2)^{3/2} = w_{vu}$, $w_{vv} = u^2/(u^2 + v^2)^{3/2}$

55. $z_{xx} = -2x/(1 + x^2)^2$, $z_{xy} = 0 = z_{yx}$, $z_{yy} = -2y/(1 + y^2)^2$

59. $12xy$, $72xy$

61. $24 \sin(4x + 3y + 2z)$, $12 \sin(4x + 3y + 2z)$

63. $\theta e^{r\theta}(2 \sin \theta + \theta \cos \theta + r \theta \sin \theta)$

65. $6yz^2$ 73. No

EXERCISES 9.3 | page 601

1. $z = -7x - 6y + 5$ 3. $x + y - 2z = 0$ 5. $z = y$

7. $2x + \frac{1}{4}y - 1$ 9. $\frac{1}{9}x - \frac{2}{9}y + \frac{2}{3}$ 11. $4x + 13y + 4z - 28$

15. 6.3 17. $\frac{3}{7}x + \frac{2}{7}y + \frac{6}{7}z$; 6.9914

19. $4T + H - 329$; 129°F 21. $\approx 0.301m - 4.89v + 96$

EXERCISES 9.4 | page 608

1. $(2x + y) \cos t + (2y + x)e^t$

3. $[(x/t) - y \sin t]/\sqrt{1 + x^2 + y^2}$

5. $e^{y/z}[2t - (x/z) - (2xy/z^2)]$

9. $\partial z/\partial s = 2xy^3 \cos t + 3x^2y^2 \sin t$, $\partial z/\partial t = -2sxy^3 \sin t + 3sx^2y^2 \cos t$

11. 62 13. $\frac{2x + y \sin x}{\cos x - 2y}$ 15. $\frac{\sin(x - y) + e^y}{\sin(x - y) - xe^y}$

17. $\frac{3yz - 2x}{2z - 3xy}$, $\frac{3xz - 2y}{2z - 3xy}$

19. $\frac{1 + y^2z^2}{1 + y + y^2z^2}$, $-\frac{z}{1 + y + y^2z^2}$

21. (a) $\frac{dW}{dt} = \frac{\partial W}{\partial F} \frac{dF}{dt} + \frac{\partial W}{\partial C} \frac{dC}{dt}$

(b) $\partial W/\partial F$ is positive, $\partial W/\partial C$ is negative.

(c) Wolf population increases.

(d) Nothing

23. $\frac{dB}{da} = \frac{1}{h^2} \frac{dm}{da} - 2 \frac{m}{h^3} \frac{dh}{da}$

25. $\frac{\partial A}{\partial p} = \frac{1}{pq - e^{qa}}$ 27. $\approx -0.27 \text{ L/s}$

EXERCISES 9.5 | page 617

1. $\approx -0.08 \text{ mb/km}$ 3. ≈ 0.778 5. $2 + \sqrt{3}/2$

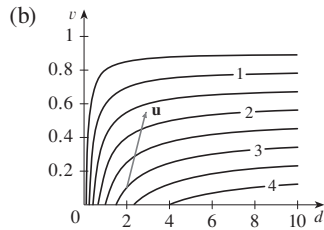
7. (a) $\nabla f(x, y) = [5y^2 - 12x^2y, 10xy - 4x^3]$

(b) $[-4, 16]$ (c) $172/13$

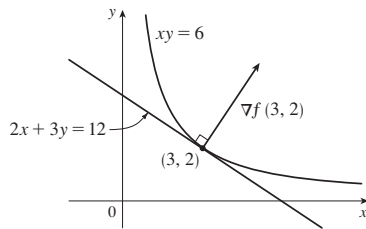
9. (a) $\nabla f(x, y) = [2 \cos(2x + 3y), 3 \cos(2x + 3y)]$

(b) $[2, 3]$ (c) $\sqrt{3} - \frac{3}{2}$

11. $23/10$ 13. $-8/\sqrt{10}$ 15. $-6/\sqrt{5}$ 17. $2/5$
 19. $4\sqrt{2}, [-1, 1]$ 21. $1, [0, 1]$
 23. (b) $[-12, 92]$ 25. All points on the line $y = x + 1$
 27. Deeper 29. $[-1, -2]$
 31. (a) It is beneficial



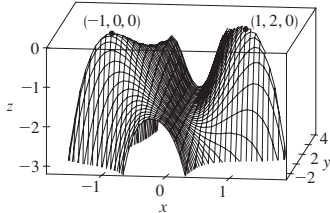
33. $[2, 3], 2x + 3y = 12$



35. $\frac{774}{25}$

EXERCISES 9.6 | page 626

1. (a) f has a local minimum at $(1, 1)$.
 (b) f has a saddle point at $(1, 1)$.
 3. Local minimum at $(1, 1)$, saddle point at $(0, 0)$
 5. Minimum $f(\frac{1}{3}, -\frac{2}{3}) = -\frac{1}{3}$
 7. Minima $f(1, 1) = 0, f(-1, -1) = 0$, saddle point at $(0, 0)$
 9. Minimum $f(2, 1) = -8$, saddle point at $(0, 0)$
 11. None 13. Minimum $f(0, 0) = 0$, saddle points at $(\pm 1, 0)$
 15. Minima $f(0, 1) = f(\pi, -1) = f(2\pi, 1) = -1$,
 saddle points at $(\pi/2, 0), (3\pi/2, 0)$
 17. Maximum $f(2, 0) = 9$, minimum $f(0, 3) = -14$
 19. Maximum $f(\pm 1, 1) = 7$, minimum $f(0, 0) = 4$
 21.



23. $\sqrt{3}$ 25. $(2, 1, \sqrt{5}), (2, 1, -\sqrt{5})$ 27. $\frac{100}{3}, \frac{100}{3}, \frac{100}{3}$
 29. $8r^3/(3\sqrt{3})$ 31. $\frac{4}{3}$ 33. Cube, edge length $c/12$
 35. Square base of side 40 cm, height 20 cm
 37. (a) $H(p_1, p_2) = \ln \left[\frac{(1 - p_1 - p_2)^{p_1 + p_2 - 1}}{p_1^{p_1} p_2^{p_2}} \right]$
 (b) $\{(p_1, p_2) \mid p_1 \geq 0, p_2 \geq 0, p_1 + p_2 \leq 1\}$
 (c) $\ln 3, p_1 = p_2 = p_3 = \frac{1}{3}$
 39. $\frac{100}{21}$

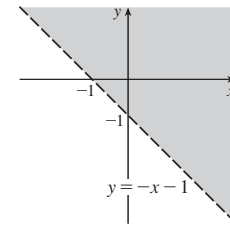
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True-False Quiz

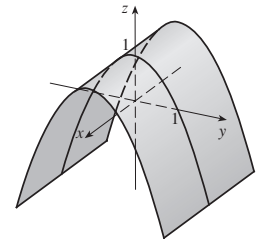
1. True 3. False 5. False 7. True 9. False
 11. True

Exercises

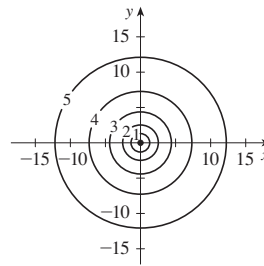
1. $\{(x, y) \mid y > -x - 1\}$



- 3.



- 5.



7. $\frac{2}{3}$

9. (a) $\approx 3.5^\circ\text{C/m}, -3.0^\circ\text{C/m}$ (b) $\approx 0.35^\circ\text{C/m}$ by Equation 9.5.9 (Definition 9.5.2 gives $\approx 1.1^\circ\text{C/m}$)

- (c) ≈ -0.25

11. $f_x = 1/\sqrt{2x + y^2}, f_y = y/\sqrt{2x + y^2}$

13. $g_u = \tan^{-1}v, g_v = u/(1 + v^2)$

15. $T_p = \ln(q + e^r), T_q = p/(q + e^r), T_r = pe^r/(q + e^r)$

17. (a) $\partial P/\partial m \approx 4.87, \partial P/\partial T \approx -0.013$

- (b) Worsen (c) Yes

19. $f_{xx} = 24x, f_{xy} = -2y = f_{yx}, f_{yy} = -2x$

21. $f_{xx} = k(k-1)x^{k-2}y^l z^m, f_{xy} = klx^{k-1}y^{l-1}z^m = f_{yx},$

- $f_{xz} = kmx^{k-1}y^l z^{m-1} = f_{zx}, f_{yy} = l(l-1)x^k y^{l-2} z^m,$

- $f_{yz} = lmx^k y^{l-1} z^{m-1} = f_{zy}, f_{zz} = m(m-1)x^k y^l z^{m-2}$

25. $z = -8x - 2y$ 27. $x - 2y + z = 4$ 29. $x + 5y - 4$

31. $2xy^3(1 + 6p) + 3x^2y^2(pe^p + e^p) + 4z^3(p \cos p + \sin p)$

35. $[q^2 e^{-pq} - pq^3 e^{-pq}, 2pq e^{-pq} - p^2 q^2 e^{-pq}]$ 37. $-\frac{4}{5}$

39. $\approx \frac{5}{8}$ knot/mi

41. Maximum $f(1, 1) = 1$; saddle points $(0, 0), (0, 3), (3, 0)$

43. Maximum $f(1, 2) = 4$, minimum $f(2, 4) = -64$

CHAPTER 10

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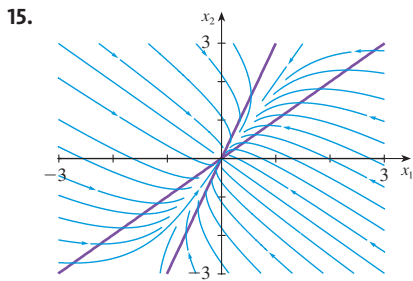
1. Linear, homogeneous, nonautonomous
 3. Nonlinear, autonomous
 5. Linear, nonhomogeneous, nonautonomous

7. $\begin{bmatrix} dx/dt \\ dy/dt \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

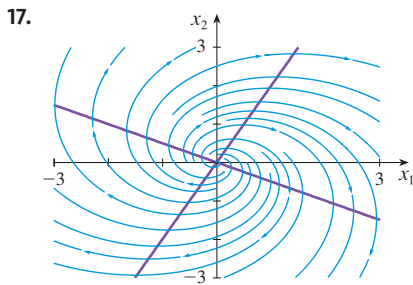
9. $\begin{bmatrix} dx/dt \\ dy/dt \end{bmatrix} = \begin{bmatrix} 0 & 3t \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -7 \\ 0 \end{bmatrix}$

11. $\begin{bmatrix} dx/dt \\ dy/dt \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -5 \\ 0 \end{bmatrix}$

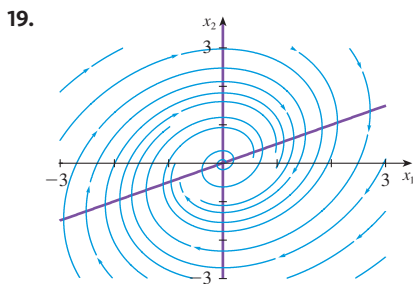
13. $\begin{bmatrix} dx/dt \\ dy/dt \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -3t \\ 0 \end{bmatrix}$



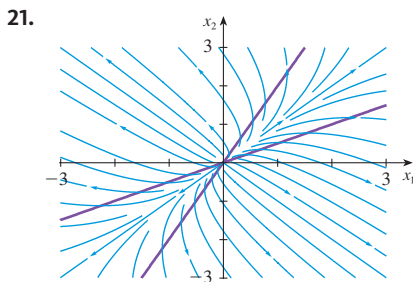
Node



Spiral



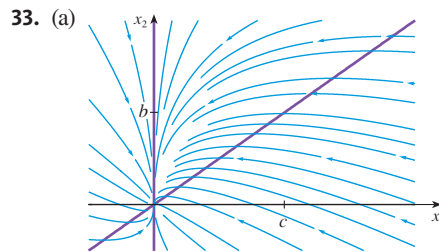
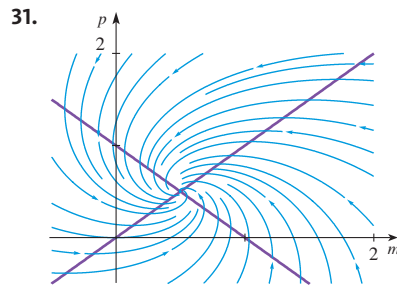
Spiral



Node

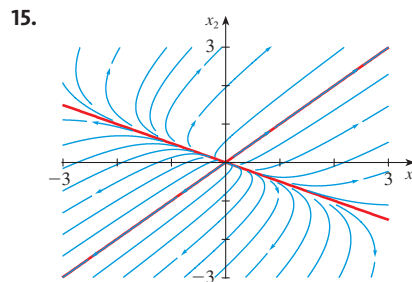
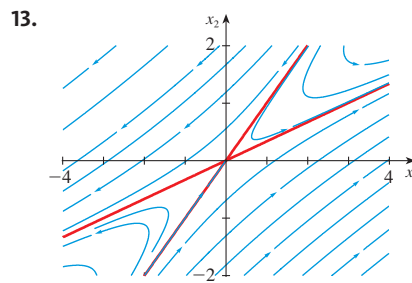
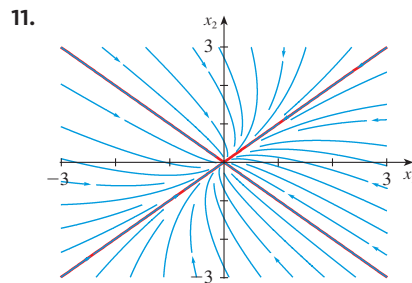
23. (a) $\hat{\mathbf{x}} = -A^{-1}\mathbf{g}$

(b) The distance between the vector $\mathbf{x}$ and the equilibrium vector $\hat{\mathbf{x}}$ along each coordinate dimension



(b) Amount of antibody in blood and tumor approaches zero.

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17. $\mathbf{x}(t) = \frac{3}{2}e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{2}e^{-2t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$19. \mathbf{x}(t) = 3e^t \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 4e^{-t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$21. \mathbf{x}(t) = -9e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 4e^{3t} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$23. \mathbf{x}(t) = e^{-t} \begin{bmatrix} 2 \cos \sqrt{6} t \\ -\sqrt{6} \sin \sqrt{6} t \end{bmatrix}$$

$$25. \mathbf{x}(t) = \frac{1}{2}e^t \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{3}{2}e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$27. \mathbf{x}(t) = e^{3t/2} \begin{bmatrix} \cos \frac{\sqrt{39}}{2} t - \frac{3\sqrt{39}}{13} \sin \frac{\sqrt{39}}{2} t \\ \cos \frac{\sqrt{39}}{2} t + \frac{\sqrt{39}}{13} \sin \frac{\sqrt{39}}{2} t \end{bmatrix}$$

$$29. \mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

31. (b) No restrictions on the components of the eigenvectors

33. $a < 0$: stable spiral; $0 \leq a < 1$: stable node;

$a = 1$: unstable (infinite number of equilibria);

$a > 1$: saddle (unstable)

35. $a < -1$: stable node;

$a = -1$: unstable (infinite number of equilibria);

$-1 < a < 1$: saddle (unstable);

$a = 1$: unstable (infinite number of equilibria);

$a > 1$: unstable node

■ EXERCISES 10.3 | page 661

$$1. \mathbf{z}(t) = e^{-5 \times 10^{-4} t} \cdot$$

$$\begin{bmatrix} k_1 \cos 0.1t + k_2 \sin 0.1t \\ (-5 \times 10^{-4} k_1 + 0.1 k_2) \cos 0.1t - (0.1 k_1 + 5 \times 10^{-4} k_2) \sin 0.1t \end{bmatrix}$$

$$3. (a) \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \text{ where } \mathbf{x}(t) = \mathbf{y}(t) - \begin{bmatrix} 0 \\ d/2 \end{bmatrix}$$

$$(b) \mathbf{x}(t) = c_1 e^{-2t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_2 e^{-4t} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$(c) \mathbf{x}(t) = (\tfrac{1}{2} - d/2) e^{-2t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \tfrac{1}{2} e^{-4t} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$(d) y_2(t) = (\tfrac{1}{2} - d/2) e^{-2t} - \tfrac{1}{2} e^{-4t} + d/2$$

$$(e) -\tfrac{1}{2} \ln \left[\tfrac{1}{2} (1 - d + \sqrt{(d-1)(d+3)}) \right]$$

$$5. (a) \mathbf{x}(t) = c_1 e^{(\rho-\delta)t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_2 e^{-(\alpha+\beta)t} \begin{bmatrix} \rho - \delta + \alpha + \beta \\ -\alpha \end{bmatrix}$$

(b) $\rho > \delta$: saddle (unstable); $\rho < \delta$: stable node

$$(c) \mathbf{x}(t) = \frac{\alpha C_0}{\rho - \delta + \alpha + \beta} e^{(\rho-\delta)t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{C_0}{\rho - \delta + \alpha + \beta} e^{-(\alpha+\beta)t} \begin{bmatrix} \rho - \delta + \alpha + \beta \\ -\alpha \end{bmatrix}$$

$$7. (a) x'_0 = -u_0 x_0, x'_1 = u_0 x_0 - u_1 x_1, x'_2 = u_1 x_1$$

$$(b) \mathbf{x}(t) = \frac{k}{u_0 - u_1} e^{-u_0 t} \begin{bmatrix} u_0 - u_1 \\ -u_0 \end{bmatrix} + \frac{u_0 k}{u_0 - u_1} e^{-u_1 t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$(c) \frac{dx_2}{dt} = -\frac{u_0 u_1 k}{u_0 - u_1} e^{-u_0 t} + \frac{u_0 u_1 k}{u_0 - u_1} e^{-u_1 t}$$

$$(d) x_2(t) = \frac{u_1 k}{u_0 - u_1} (e^{-u_0 t} - 1) - \frac{u_0 k}{u_0 - u_1} (e^{-u_1 t} - 1)$$

9. (a) Water temperature affected only by room temperature; coin temperature affected only by water temperature

$$(b) \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \text{ where } \mathbf{x} = \begin{bmatrix} w(t) - R \\ p(t) - R \end{bmatrix}$$

$$(c) \mathbf{x}(t) = c_1 e^{-k_w t} \begin{bmatrix} k_w - k_p \\ -k_p \end{bmatrix} + c_2 e^{-k_p t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$(d) \begin{bmatrix} w(t) \\ p(t) \end{bmatrix} = \frac{w_0 - R}{k_w - k_p} e^{-k_w t} \begin{bmatrix} k_w - k_p \\ -k_p \end{bmatrix} + \left(k_p \frac{w_0 - R}{k_w - k_p} + p_0 - R \right) e^{-k_p t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} R \\ R \end{bmatrix}$$

$$11. (a) \hat{x}_1 = \hat{x}_2 = cV$$

$$(b) \frac{d\mathbf{y}}{dt} = \mathbf{A}\mathbf{y}, \text{ where } \mathbf{y} = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} - \begin{bmatrix} cV/2 \\ 0 \end{bmatrix}$$

$$(c) \mathbf{y}(t) = cV e^{-at/V} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{cV}{2} e^{-2at/V} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(d) \mathbf{x}(t) = cV e^{-at/V} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{cV}{2} e^{-2at/V} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} cV/2 \\ 0 \end{bmatrix}$$

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$$1. \frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} 4 & 0 \\ 0 & -2 \end{bmatrix} \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$3. \frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$5. \frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$7. (i) \hat{x}_1 = 0, \hat{x}_2 = 0; \frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} -5 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(ii) \hat{x}_1 = \tfrac{1}{5}, \hat{x}_2 = 5; \frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} 0 & \tfrac{1}{5} \\ -25 & 0 \end{bmatrix} \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} \tfrac{1}{5} \\ 5 \end{bmatrix}$$

$$9. (i) \hat{x}_1 = 0, \hat{x}_2 = 0; \frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} 1 & 0 \\ 8 & 0 \end{bmatrix} \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(ii) \hat{x}_1 = -24, \hat{x}_2 = -2;$$

$$\frac{d\boldsymbol{\epsilon}}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -96 \end{bmatrix} \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} -24 \\ -2 \end{bmatrix}$$

11. (i) $\hat{x}_1 = 0, \hat{x}_2 = 0; \frac{d\mathbf{\epsilon}}{dt} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \mathbf{\epsilon}$, where $\mathbf{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(ii) $\hat{x}_1 = -1, \hat{x}_2 = -1;$

$\frac{d\mathbf{\epsilon}}{dt} = \begin{bmatrix} e & -e \\ -1 & 0 \end{bmatrix} \mathbf{\epsilon}$, where $\mathbf{\epsilon} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

13. (i) unstable, (ii) locally stable

15. (i) inconclusive, (ii) unstable

17. (i) locally stable, (ii) unstable

19. (i) $\hat{x} = 0, \hat{y} = 0$; unstable (ii) $\hat{x} = 0, \hat{y} = 2$; unstable

(iii) $\hat{x} = 3, \hat{y} = 0$; locally stable

21. (i) $\hat{n} = 0, \hat{m} = 0$; unstable

(ii) $\hat{n} = 0, \hat{m} = 2$; locally stable

(iii) $\hat{n} = \frac{3}{4}, \hat{m} = \frac{1}{2}$; unstable

23. (i) $\hat{p} = -\frac{1}{2} + \sqrt{5}/2, \hat{q} = \frac{5}{2} - \sqrt{5}/2$; locally stable

(ii) $\hat{p} = -\frac{1}{2} - \sqrt{5}/2, \hat{q} = \frac{5}{2} + \sqrt{5}/2$; unstable

25. (i) $\hat{x} = 0, \hat{y} = 0$; locally stable when $a < 1$

(ii) $\hat{x} = \frac{1-a}{a}, \hat{y} = \frac{1-a}{a}$; locally stable when $a > 1$

27. (a) (i) $\hat{p}_1 = 0, \hat{p}_2 = 0$, (ii) $\hat{p}_1 = 0, \hat{p}_2 = \frac{9}{10}$,

(iii) $\hat{p}_1 = \frac{2}{5}, \hat{p}_2 = 0$, and (iv) $\hat{p}_1 = \frac{2}{5}, \hat{p}_2 = \frac{13}{30}$

(b) $J(p_1, p_2) = \begin{bmatrix} 2 - 10p_1 & 0 \\ -35p_2 & 27 - 35p_1 - 60p_2 \end{bmatrix}$

(c) (i), (ii), and (iii) are unstable; (iv) is locally stable.

(d) Yes

29. $\hat{R} = 1, \hat{C} = 2$; locally stable

31. (i) $\hat{R} = 0, \hat{C} = 0$; unstable

(ii) $\hat{R} = 5, \hat{C} = 0$; unstable

(iii) $\hat{R} = 1, \hat{C} = \frac{8}{5}$; locally stable

33. (a) dx/dt and dy/dt do not contain any terms with z

(b) $\hat{x} = 0, \hat{y} = 1$

(c) $J(x, y) = \begin{bmatrix} -k_f y M & -M(k_f x + k_r) \\ -k_f y M & -M(k_f x + k_r + k_{cat}) \end{bmatrix}$

(d) Locally stable

35. (b) $J(v, w) = \begin{bmatrix} -3v^2 + 2(1+a)v - a & -1 \\ \epsilon & -\epsilon \end{bmatrix}$

(c) Locally stable for all positive constants a and ϵ

13. (a) $\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{4}{25} & 0 \\ \frac{4}{25} & -\frac{8}{25} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 16 \\ 0 \end{bmatrix}$ and $\mathbf{x}(0) = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$

(b) $\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = -92e^{-4t/25} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 42e^{-8t/25} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 100 \\ 50 \end{bmatrix}$

(c) 50 g

19. Cannot tell 21. Locally stable

23. (i) $\hat{x} = 0, \hat{y} = 0$, unstable

(ii) $\hat{x} = 0, \hat{y} = 4$, unstable

(iii) $\hat{x} = -\frac{1}{2}, \hat{y} = 5$, locally stable

25. (i) $\hat{R} = 0, \hat{W} = 0$, unstable

(ii) $\hat{R} = \frac{1}{2}, \hat{W} = 0$, locally stable

(iii) $\hat{R} = 1, \hat{W} = -8$, unstable

27. (a) (i) $\hat{N}_1 = 0, \hat{N}_2 = 0$, (ii) $\hat{N}_1 = 0, \hat{N}_2 = K_2$,

(iii) $\hat{N}_1 = K_1, \hat{N}_2 = 0$, (iv) $\hat{N}_1 = \frac{\alpha K_2 - K_1}{\alpha\beta - 1}, \hat{N}_2 = \frac{K_1\beta - K_2}{\alpha\beta - 1}$

(b) $J(N_1, N_2) = \begin{bmatrix} 1 - \frac{2N_1 + \alpha N_2}{K_1} & -\frac{\alpha N_1}{K_1} \\ -\frac{\beta N_2}{K_2} & 1 - \frac{2N_2 + \beta N_1}{K_2} \end{bmatrix}$

(c) (i) and (ii) are unstable; (iii) is locally stable; (iv) is not biologically feasible

(d) (i) and (iii) are unstable; (ii) is locally stable; (iv) is not biologically feasible

(e) (i) and (iv) are unstable; (ii) and (iii) are locally stable

(f) (i), (ii), and (iii) are unstable; (iv) is locally stable

29. $\frac{dp_1}{dt} = f_1(p_1, p_2) = c_1 p_1 (h - p_1) - m_1 p_1;$

$\frac{dp_2}{dt} = f_2(p_1, p_2) = c_2 p_2 (h - p_1 - p_2) - m_2 p_2 - c_1 p_1 p_2$

(a) (i) $\hat{p}_1 = 0, \hat{p}_2 = 0$, (ii) $\hat{p}_1 = 0, \hat{p}_2 = h - \frac{m_2}{c_2}$,

(iii) $\hat{p}_1 = h - \frac{m_1}{c_1}, \hat{p}_2 = 0$, and

(iv) $\hat{p}_1 = h - \frac{m_1}{c_1}, \hat{p}_2 = \frac{m_1}{c_1} - \frac{m_2}{c_2} - \frac{c_1 h}{c_2} + \frac{m_1}{c_2}$

(b)

$J(p_1, p_2) = \begin{bmatrix} c_1(h - 2p_1) - m_1 & 0 \\ -c_2 p_2 - c_1 p_2 & c_2(h - p_1 - 2p_2) - m_2 - c_1 p_1 \end{bmatrix}$

(c) $hc_1 < m_1$ and $hc_2 < m_2$

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True-False Quiz

1. False 3. True 5. True 7. True 9. False

Exercises

1. Nonlinear 3. Linear

9. $\mathbf{x}(t) = \begin{bmatrix} \cos t + \sin t \\ \cos t - \sin t \end{bmatrix}$

11. $\mathbf{x}(t) = \frac{1}{3}e^t \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \frac{2}{3}e^{-2t} \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

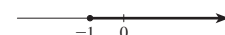
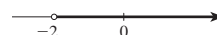
APPENDIXES

APPENDIX A | page 688

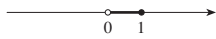
1. 18 3. $5 - \sqrt{5}$ 5. $2 - x$

7. $|x + 1| = \begin{cases} x + 1 & \text{for } x \geq -1 \\ -x - 1 & \text{for } x < -1 \end{cases}$ 9. $x^2 + 1$

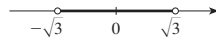
11. $(-2, \infty)$ 13. $[-1, \infty)$



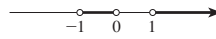
15. $(0, 1]$



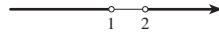
19. $(-\sqrt{3}, \sqrt{3})$



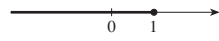
23. $(-1, 0) \cup (1, \infty)$



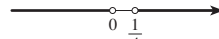
17. $(-\infty, 1) \cup (2, \infty)$



21. $(-\infty, 1]$



25. $(-\infty, 0) \cup (\frac{1}{4}, \infty)$



27. $10 \leq C \leq 35$

29. (a) $T = 20 - 10h, 0 \leq h \leq 12$

(b) $-30^\circ\text{C} \leq T \leq 20^\circ\text{C}$

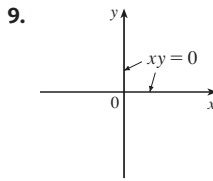
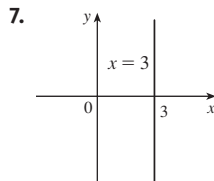
31. $2, -\frac{4}{3}$ 33. $(-3, 3)$

35. $(3, 5)$ 37. $(-\infty, -7] \cup [-3, \infty)$ 39. $[1.3, 1.7]$

41. $x \geq (a + b)c/(ab)$

■ APPENDIX B | page 698

1. 5 3. $-\frac{9}{2}$

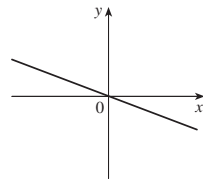


11. $y = 6x - 15$ 13. $5x + y = 11$ 15. $y = 3x - 2$

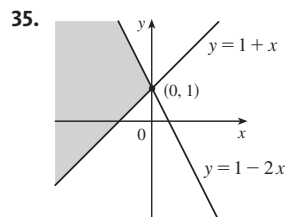
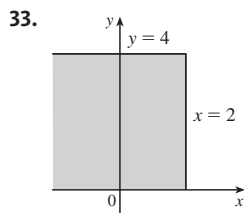
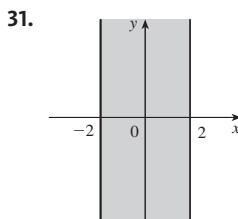
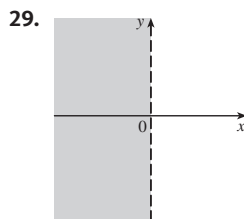
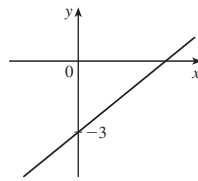
17. $y = 3x - 3$ 19. $y = 5$ 21. $x + 2y + 11 = 0$

23. $5x - 2y + 1 = 0$

25. $m = -\frac{1}{3}, b = 0$

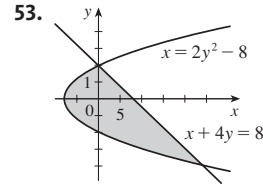


27. $m = \frac{3}{4}, b = -3$



37. $(x - 3)^2 + (y + 1)^2 = 25$ 39. $(2, -5), 4$ 41. $(1, -2)$

45. $y = x - 3$

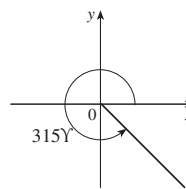


■ APPENDIX C | page 707

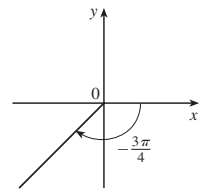
1. (a) $7\pi/6$ (b) $\pi/20$ 3. (a) 720° (b) -67.5°

5. 3π cm 7. $\frac{2}{3}$ rad $= (120/\pi)^\circ$

9. (a)



(b)



11. $\sin(3\pi/4) = 1/\sqrt{2}, \cos(3\pi/4) = -1/\sqrt{2}, \tan(3\pi/4) = -1,$
 $\csc(3\pi/4) = \sqrt{2}, \sec(3\pi/4) = -\sqrt{2}, \cot(3\pi/4) = -1$

13. $\cos \theta = \frac{4}{5}, \tan \theta = \frac{3}{4}, \csc \theta = \frac{5}{3}, \sec \theta = \frac{5}{4}, \cot \theta = \frac{4}{3}$

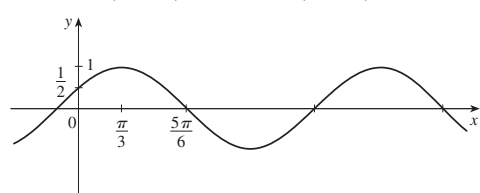
15. 5.73576 cm 17. 24.62147 cm 29. $\frac{1}{15}(4 + 6\sqrt{2})$

31. $\pi/3, 5\pi/3$ 33. $\pi/6, \pi/2, 5\pi/6, 3\pi/2$

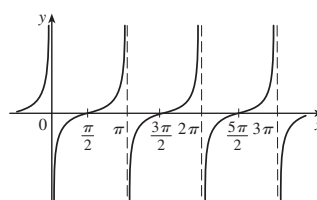
35. $0 \leq x \leq \pi/6$ and $5\pi/6 \leq x \leq 2\pi$

37. $0 \leq x < \pi/4, 3\pi/4 < x < 5\pi/4, 7\pi/4 < x \leq 2\pi$

39.



41.



■ APPENDIX D | page 713

1. (a) $n > 100$

3. $\frac{4}{7}$ (or any smaller positive number)

5. 1.44 (or any smaller positive number)

7. 0.0906 (or any smaller positive number)

9. 0.11 or 0.012 (or smaller positive numbers)

13. (a) $\sqrt{1000/\pi}$ cm (b) Within approximately 0.0445 cm

(c) Radius; area; $\sqrt{1000/\pi}$; 1000; 5; ≈ 0.0445

15. (a) 0.025 (b) 0.0025

■ APPENDIX F | page 722

1. $\sqrt{1} + \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{5}$ 3. $3^4 + 3^5 + 3^6$

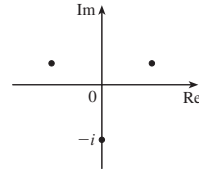
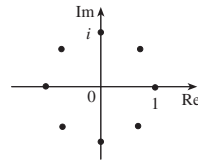
5. $-1 + \frac{1}{3} + \frac{3}{5} + \frac{5}{7} + \frac{7}{9}$ 7. $1^{10} + 2^{10} + 3^{10} + \dots + n^{10}$

9. $1 - 1 + 1 - 1 + \cdots + (-1)^{n-1}$ 11. $\sum_{i=1}^{10} i$
 13. $\sum_{i=1}^{19} \frac{i}{i+1}$ 15. $\sum_{i=1}^n 2i$ 17. $\sum_{i=0}^5 2^i$ 19. $\sum_{i=1}^n x^i$
 21. 80 23. 3276 25. 0 27. 61 29. $n(n+1)$
 31. $n(n^2 + 6n + 17)/3$ 33. $n(n^2 + 6n + 11)/3$
 35. $n(n^3 + 2n^2 - n - 10)/4$
 41. (a) n^4 (b) $5^{100} - 1$ (c) $\frac{97}{300}$ (d) $a_n - a_0$
 43. $\frac{1}{3}$ 45. 14 49. $2^{n+1} + n^2 + n - 2$

■ APPENDIX G | page 731

1. $8 - 4i$ 3. $13 + 18i$ 5. $12 - 7i$ 7. $\frac{11}{13} + \frac{10}{13}i$
 9. $\frac{1}{2} - \frac{1}{2}i$ 11. $-i$ 13. $5i$ 15. $12 + 5i, 13$
 17. $4i, 4$ 19. $\pm \frac{3}{2}i$ 21. $-1 \pm 2i$
 23. $-\frac{1}{2} \pm (\sqrt{7}/2)i$ 25. $3\sqrt{2} [\cos(3\pi/4) + i \sin(3\pi/4)]$
 27. $5\{\cos[\tan^{-1}(\frac{4}{3})] + i \sin[\tan^{-1}(\frac{4}{3})]\}$

29. $4[\cos(\pi/2) + i \sin(\pi/2)], \cos(-\pi/6) + i \sin(-\pi/6),$
 $\frac{1}{2}[\cos(-\pi/6) + i \sin(-\pi/6)]$
 31. $4\sqrt{2} [\cos(7\pi/12) + i \sin(7\pi/12)],$
 $(2\sqrt{2})[\cos(13\pi/12) + i \sin(13\pi/12)], \frac{1}{4}[\cos(\pi/6) + i \sin(\pi/6)]$
 33. -1024 35. $-512\sqrt{3} + 512i$
 37. $\pm 1, \pm i, (1/\sqrt{2})(\pm 1 \pm i)$ 39. $\pm(\sqrt{3}/2) + \frac{1}{2}i, -i$



41. i 43. $\frac{1}{2} + (\sqrt{3}/2)i$ 45. $-e^2$
 47. $\cos 3\theta = \cos^3\theta - 3 \cos \theta \sin^2\theta,$
 $\sin 3\theta = 3 \cos^2\theta \sin \theta - \sin^3\theta$

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